Statistical Matching via Schrödinger Bridge beyond Conditional Independence

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Abstract

Statistical matching combines partially overlapping datasets that share covariates X but observe the target Y and auxiliary variables Z separately. Classical approaches typically invoke the conditional independence assumption (CIA), which makes the problem identifiable but fundamentally implies that the imported auxiliary variable provides no additional predictive power for Y once X is known. To capture this latent Y–Z dependence, we propose a novel dependency-aware Schrödinger bridge for predictive statistical matching. Our approach couples the two separated databases by tilting the conservative CIA baseline with a transportation-based compatibility cost, recovering an informative joint distribution. The resulting statistical learning framework yields full probabilistic posterior rules for bidirectional imputation. Theoretically, we establish a sufficient condition under which the learned bridge strictly improves over the CIA baseline, alongside an exact joint recovery guarantee in the Gaussian setting under an appropriate cost. Across synthetic benchmarks and real-world datasets, our method improves downstream predictive utility when the compatibility cost captures informative residual dependence. These findings motivate applications such as data recoding.

1 Introduction

Integrating disparate databases is a ubiquitous technique across many real-world applications. For instance, Lee et al. (2025) demonstrate that merging supermarket loyalty data with credit card records reveals grocery shopping habits as a powerful proxy for creditworthiness, significantly improving risk predictions for consumers lacking traditional credit scores. Beyond finance, data integration is pivotal in survey studies (Jauslin and Tillé, 2023), data recoding (Garès and Omer, 2022), combining multi-assay genomics (Cai et al., 2016), multimodal learning (Li et al., 2023), and socio-economic analyses of income and poverty (Serafino and Tonkin, 2017).

Statistical matching (SM) becomes essential when these files must be merged but lack unique identification keys, often due to strict privacy constraints. Formally, let $P_{X,Y}$ and $P_{X,Z}$ be the observed marginals of an unknown joint population $P_{X,Y,Z}$ over a random vector $(X, Y, Z) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$. We consider a recipient database $D^A = \{(X_i^A, Y_i^A)\}_{i=1}^{n_A} \sim P_{X,Y}$ and a donor database $D^B = \{(X_j^B, Z_j^B)\}_{j=1}^{n_B} \sim P_{X,Z}$. Here, the covariates X are shared, but Y is exclusive to D^A and Z is exclusive to D^B. A concrete application is data recoding, where two surveys record the same

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underlying attribute using different labeling schemes (Garès and Omer, 2022). For example, Y and Z may represent alternative occupational classifications, with shared demographic covariates X providing a basis for imputing the unobserved classification in each survey. With a slight abuse of notation, we write $D^A = (X^A, Y^A)$ and $D^B = (X^B, Z^B)$. The goal of SM is to impute the missing auxiliary variables $\hat{Z}^A$ to construct a synthetic joint dataset $\hat{D}^A = (X^A, Y^A, \hat{Z}^A)$ that faithfully approximates the true joint law $P_{X,Y,Z}$. Symmetrically, D^A can act as a donor to complete $\hat{D}^B = (X^B, \hat{Y}^B, Z^B)$, yielding a fully merged pooled dataset $\hat{D}^T = \hat{D}^A \cup \hat{D}^B$.

The two observed marginals alone are generally insufficient to identify the full joint distribution. To make this joint-recovery problem tractable, classical SM (Rubin, 1986; Aluja-Banet et al., 2007; Endres and Augustin, 2016; Annoye et al., 2024) typically relies on the conditional independence assumption (CIA), $Y \perp Z \mid X$, to transfer Z^B to the recipient file. While CIA converts an ill-posed identification problem into a feasible imputation task, it is fundamentally restrictive for downstream applications. Under CIA, $P_{Y|X,Z} = P_{Y|X}$, meaning that the imported variable $\hat{Z}^A$ provides absolutely no additional predictive power for Y once X is known. Consequently, standard CIA-based integration is superfluous if the goal is to capture meaningful Y–Z interactions. Dependency-aware data integration therefore requires a principled mechanism to impose structural conditions that recover plausible, informative associations among (X, Y, Z) beyond mere conditional independence.

This work introduces a novel structurally guided SM method beyond CIA by leveraging the Schrödinger bridge mechanism (Léonard, 2014; Chen et al., 2021). Rooted in entropic optimal transport, we formulate a bridge problem that synthesizes a joint distribution $Q^\star$ whose marginals exactly match the observed $P_{X,Y}$ and $P_{X,Z}$. The constructed dependency of $Q^\star$ is anchored to the conservative CIA product coupling $P_\rho = P_X P_{Y|X} P_{Z|X}$, but is exponentially tilted by a compatibility cost to recover the latent association between D^A and D^B that minimizes distributional discrepancy. Finally, the learned posterior $Q^\star_{Z|X,Y}$ serves as a generative model for imputing $\hat{Z}^A$ conditionally on D^A. Because this approach yields a full probabilistic posterior rather than deterministic point matches, it inherently supports stochastic completion and multiple imputation (Rubin, 1987) for principled uncertainty quantification.

Contributions. Our contributions are fourfold. First, to our knowledge, this is the first work to cast statistical matching as a dependency-aware Schrödinger bridge, providing a principled mechanism to relax the restrictive conditional independence assumption (CIA). Second, we establish a theoretical sufficient condition under which the proposed bridge strictly improves recipient-side posterior completion over the CIA baseline, and we prove that the optimal bridge $Q^\star$ exactly recovers the full joint law under Gaussianity for an appropriate choice of the regularization parameter. Third, we develop a scalable algorithmic framework to estimate the bridge, transforming the potential equations into an efficient statistical risk minimization problem parameterized by neural networks. Finally, across Gaussian, nonlinear synthetic, and real-world benchmark datasets, we empirically demonstrate that our learned couplings can consistently improve downstream predictive utility compared to classical mixed methods and a recent state-of-the-art optimal transport baseline, when a small amount of auxiliary information about the population is available.

2 Schrödinger Bridge for Statistical Matching

2.1 Local and global bridge formulations

For any $x \in \mathcal{X}$, let $\rho_x := P_{Y|X=x} \otimes P_{Z|X=x}$ denote the independent product coupling induced by the CIA. To reconstruct a conditional distribution that captures the latent Y–Z dependence beyond ρ_x, we propose solving an entropically regularized optimal transport (EOT) problem for each $x \in \mathcal{X}$:

$$\pi_x^\star \in \arg\min_{\pi_x \in \Pi(P_{Y|X=x},\, P_{Z|X=x})} \left\{ \mathbb{E}_{\pi_x}[c(x, Y, Z)] + \lambda \, \mathrm{KL}(\pi_x \| \rho_x) \right\}, \tag{1}$$

where $\mathrm{KL}(\mu\|\nu) = \int \log(d\mu/d\nu) d\mu$ is the KL-divergence, and $\Pi(\mu, \nu)$ is the set of joint laws with given marginals μ and ν. The function $c : \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \to [0, \infty)$ acts as a compatibility cost, where smaller values indicate stronger latent association between x, y, and z. This objective naturally balances two forces: the first term favors couplings that place mass on compatible (y, z) pairs at the given covariate value x, while the second penalizes deviation from the conservative CIA baseline ρ_x. Thus, a small λ emphasizes compatibility, whereas a large λ keeps the solution close to independence.

To interpret (1) from a Schrödinger bridge perspective, we define the tilted reference distribution $dR_x(y,z) = \alpha(x)^{-1}\exp\{-c(x,y,z)/\lambda\}\,d\rho_x(y,z)$, where $\alpha(x)$ is the normalizing constant. Through this exponential tilt, R_x is no longer a product measure; it encodes conditional dependence between Y and Z while remaining anchored to ρ_x. For any coupling $\pi_x \in \Pi(P_{Y|X=x}, P_{Z|X=x})$, a standard calculation gives $\mathrm{KL}(\pi_x\|R_x) = \mathrm{KL}(\pi_x\|\rho_x) + \frac{1}{\lambda}\,\mathbb{E}_{\pi_x}[c(x,Y,Z)] + \log\alpha(x)$. Hence, minimizing $\mathrm{KL}(\pi_x\|R_x)$ over couplings with fixed marginals is equivalent to solving (1). We refer to Léonard (2014) and Nutz (2022) for a broader background on Schrödinger bridges and EOT.

Solving (1) for every x is conceptually useful but computationally prohibitive when the covariate space $\mathcal{X}$ is large or continuous, which is common in real-world applications. To address this, we lift the local bridge into a single global formulation. Let $P_\rho(dx,dy,dz) = P_X(dx)P_{Y|X}(dy|x)P_{Z|X}(dz|x)$ denote the full joint law under CIA, and define a single global reference measure:

$$dR(x,y,z) \propto \exp\{-c(x,y,z)/\lambda\}\,dP_\rho(x,y,z).$$

We then define the global Schrödinger bridge $Q^\star$ as the solution to:

$$Q^\star \in \arg\min_{Q\in\Pi(P_{X,Y},\,P_{X,Z})} \mathrm{KL}(Q\|R), \tag{2}$$

where $\Pi(P_{X,Y}, P_{X,Z})$ is the set of joint laws on (X,Y,Z) whose (X,Y)- and (X,Z)-marginals match the observed data. Because any candidate Q and the reference R share the same marginal P_X, a standard decomposition of the KL divergence ensures the local and global formulations are equivalent: $Q^\star$ solves (2) if and only if its conditional kernel $Q^\star_{Y,Z|X=x}$ solves (1) for P_X-almost every x. Thus, the global formulation (2) provides a single, scalable, and tractable objective for practical implementation.

2.1.1 Structural comparison to existing SOTA model

Our Schrödinger bridge formulation recovers, in the unregularized limit, the recent optimal transport approach of Garès and Omer (2022), hereafter GO. To synthesize the association between D^A and D^B on a finite support, GO constructs a joint coupling $\gamma^\star \in \arg\min_{\gamma\in\Pi(P_{X,Y},P_{X,Z})}\int c(y,z)\,d\gamma$, where the cost $c(y,z) := \mathbb{E}[d(X^A,X^B) \mid Y^A = y, Z^B = z]$ reflects the expected covariate discrepancy. This strategy aligns the datasets by minimizing the prespecified matching loss. Conceptually, our local bridge in (1) reduces to GO's unregularized transport problem as $\lambda \to 0$. However, our entropically regularized formulation offers several distinct structural and computational advantages:

• **Strict Convexity and Stability:** GO requires solving a linear program, which can suffer from instability and non-unique solutions. In contrast, the KL-penalty in (1) makes our optimization strictly convex. As established in the EOT literature, this regularization dramatically improves both statistical stability and computational efficiency (Cuturi, 2013; Peyré and Cuturi, 2019).

• **Universal support:** The standard GO formulation is strictly tailored to discrete settings and relies on tabular linear programming. Because our bridge is formulated through continuous reference measures, it naturally accommodates both continuous and discrete (one-hot encoded) variables, making it highly scalable to complex real-world predictive pipelines.

While the idea of optimal transport in statistical matching has been explored previously (Kadane, 2001; Jauslin and Tillé, 2023; Garès and Omer, 2022), to our knowledge, this work represents the first attempt to cast statistical matching as a dependency-aware Schrödinger bridge.

2.2 Posterior completion

Assuming the relevant distributions admit densities with respect to an underlying base measure, let $q^\star$, $p_{X,Y}$, and $p_{X,Z}$ denote the densities of $Q^\star$, $P_{X,Y}$, and $P_{X,Z}$, respectively. The solution to the global formulation in (2) admits a factorized representation characterized by two potential functions.

Proposition 2.1. *The density $q^\star$ of the solution $Q^\star$ to* (2) *factors as*

$$q^\star(x,y,z) = A(x,y)B(x,z)e^{-c(x,y,z)/\lambda}, \tag{3}$$

where the positive potentials $A(x,y)$ and $B(x,z)$ are determined by the system of equations:

$$A(x,y)\int B(x,z)e^{-c(x,y,z)/\lambda}dz = p_{X,Y}(x,y),\;\; B(x,z)\int A(x,y)e^{-c(x,y,z)/\lambda}dy = p_{X,Z}(x,z). \tag{4}$$

Consequently, the posterior conditional densities for completion are given by

$$q^{\star}_{Z|X,Y}(z \mid x, y) \propto B(x,z)e^{-c(x,y,z)/\lambda}, \quad q^{\star}_{Y|X,Z}(y \mid x, z) \propto A(x,y)e^{-c(x,y,z)/\lambda}. \tag{5}$$

This representation directly yields a stochastic completion rule. For each recipient unit $i = 1, \ldots, n_A$, we impute the missing auxiliary variable by sampling from the learned posterior:

$$\hat{Z}_i^A \sim q^{\star}_{Z|X,Y}(\cdot \mid X_i^A, Y_i^A) \propto B(X_i^A, z)e^{-c(X_i^A, Y_i^A, z)/\lambda}, \tag{6}$$

yielding the completed dataset $\hat{D}^A$. By drawing from a generative posterior rather than deterministic point matches, each sampling pass yields a distinct completed database. This stochasticity provides the functionality of multiple imputation (Rubin, 1987), allowing downstream models to properly quantify matching uncertainty. The completion for $\hat{D}^B$ proceeds symmetrically using $q^{\star}_{Y|X,Z}$.

2.3 A learning framework to find the potentials

We propose a scalable statistical learning framework to compute the bridge potentials. The central idea is to transform the recursive integral equations in Proposition 2.1 into a statistical risk minimization problem. Let $A_p(x,y) = A(x,y)/p_{X,Y}(x,y)$. As the two observed datasets share the same marginal distribution P_X, the ratio of their joint densities simplifies to $d(x,y,z) := p_{X,Y}(x,y)/p_{X,Z}(x,z) = p_{Y|X}(y \mid x)/p_{Z|X}(z \mid x)$. We can thus rewrite the system (4) as

$$A_p(x,y) \int B(x,z)e^{-c(x,y,z)/\lambda}dz = 1, \;\; B(x,z) \int A_p(x,y)d(x,y,z)e^{-c(x,y,z)/\lambda}dy = 1. \tag{7}$$

We introduce an auxiliary proposal distribution q (e.g., a simple base measure or empirical marginal) for Y and Z. By importance weighting, the conditions in (7) are equivalent to:

$$A_p(x,y)\mathbb{E}_{Z\sim q}\left[\frac{B(x,Z)}{q(Z)}e^{-c(x,y,Z)/\lambda}\right] = 1, \;\; B(x,z)\mathbb{E}_{Y\sim q}\left[\frac{A_p(x,Y)}{q(Y)}d(x,Y,z)e^{-c(x,Y,z)/\lambda}\right] = 1.$$

We parameterize A_p and B using sufficiently flexible neural networks $A_p(x,y;\theta)$ and $B(x,z;\phi)$. To control the overall scale of the potentials, we constrain the potential B to have unit norm, $\|B(\cdot;\phi)\|_{L^2(P_{X,Z})} = 1$, via normalization. We can now define a statistical risk for learning the optimal parameters θ and ϕ:

$$\begin{aligned} L(\theta,\phi) = {} & \mathbb{E}_{P_{X,Y}}\big[\big\{A_p(X,Y;\theta)\mathbb{E}_{Z\sim q}\big[\big(B(X,Z;\phi)/q(Z)\big)e^{-c(X,Y,Z)/\lambda}\big] - 1\big\}^2\big] \\ & + \mathbb{E}_{P_{X,Z}}\big[\big\{B(X,Z;\phi)\mathbb{E}_{Y\sim q}\big[\big(A_p(X,Y;\theta)/q(Y)\big)d(X,Y,Z)e^{-c(X,Y,Z)/\lambda}\big] - 1\big\}^2\big]. \end{aligned} \tag{8}$$

By construction, $L(\theta,\phi) \geq 0$, and the optimal parameters (θ^*, ϕ^*) that achieve $L(\theta^*, \phi^*) = 0$ satisfy the system (7) almost surely. In practice, we minimize the empirical estimate of $L(\theta,\phi)$ using alternating gradient descent. The outer expectations are approximated using mini-batches from D^A and D^B, while the inner expectations are estimated via Monte Carlo sampling from q. The density ratio $d(x,y,z)$ can be readily evaluated by pre-fitting conditional density estimators using off-the-shelf methods. Finally, the learned potentials $A_p(x,y;\theta^*)$ and $B(x,z;\phi^*)$ are used to execute the posterior completion. For instance, if the support of Z is finite (or empirically approximated by the donor pool $\hat{\mathcal{Z}} = \{Z_j \in D^B\}$), the imputation procedure in (6) reduces to sampling from a categorical distribution where the probability of selecting $z_k \in \hat{\mathcal{Z}}$ is proportional to $B(x,z_k;\phi^*)e^{-c(x,y,z_k)/\lambda}$.

3 Theoretical Properties

We now investigate when the bridge-based completion strictly improves upon the CIA baseline, and when it is capable of exact joint recovery. Recall that $P = P_{X,Y,Z}$ denotes the true joint law and $P_\rho = P_X P_{Y|X} P_{Z|X}$ denotes the product law induced by CIA.

3.1 A sufficient condition for informative improvement

For any candidate joint law Q, we quantify the recipient-side predictive gain over the CIA baseline by the expected difference in Kullback-Leibler divergences:

$$\Delta_Z(Q) = \mathbb{E}_{P_{X,Y}}\big[\mathrm{KL}(P_{Z|X,Y} \,\|\, P_{Z|X}) - \mathrm{KL}(P_{Z|X,Y} \,\|\, Q_{Z|X,Y})\big].$$

A strictly positive $\Delta_Z(Q) > 0$ indicates that $Q_{Z|X,Y}$ provides a more accurate posterior for recipient-side completion on average than the conservative baseline $P_{Z|X}$.

Theorem 3.1. *Assume the cost satisfies $\mathbb{E}_{P_\rho}[c(X,Y,Z)^2] < \infty$ and $c \geq 0$. Define the constants*

$$\delta_c = (\mathbb{E}_{P_\rho} - \mathbb{E}_P)[c(X,Y,Z)], \;\; C = \frac{1}{2}\mathbb{E}_{P_\rho}[c(X,Y,Z)^2], \;\; C_\lambda = \frac{1}{2}\left(\mathbb{E}_{P_\rho}[e^{-c(X,Y,Z)/\lambda}] - 1\right)^2.$$

Let $Q^\star$ be the solution to the global bridge in (2)*. Then, for every $\lambda > 0$, the informative gain satisfies*

$$\Delta_Z(Q^\star) \geq \frac{\delta_c}{\lambda} - \frac{C}{\lambda^2} + C_\lambda. \tag{9}$$

Theorem 3.1 identifies the expected cost difference δ_c as the primary driver of improvement. Specifically, if $\delta_c > 0$—meaning the cost successfully assigns smaller penalties to (x, y, z) triples that are more probable under the true dependence than under the independent baseline—the leading term δ_c/λ dominates, guaranteeing a strictly positive predictive gain $\Delta_Z(Q^\star) > 0$ for appropriately large λ. By symmetry, an identical conclusion holds for donor-side completion via $Q^\star_{Y|X,Z}$.

3.2 Joint recovery for Gaussian distributions

We now analyze the conditions for exact joint recovery, focusing on the case where $P_{X,Y,Z}$ is a multivariate Gaussian distribution with an x-independent compatibility cost, i.e., $c(x, y, z) = c(y, z)$.

Theorem 3.2. *Suppose $X \in \mathbb{R}^d$, $Y \in \mathbb{R}$, and $Z \in \mathbb{R}$ are jointly Gaussian with law $P = \mathcal{N}(m, \Sigma)$. Let $\sigma_{YZ|X} = \Sigma_{YZ} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XZ}$ denote the true conditional covariance between Y and Z given X. Consider the cost functions*

$$c_{\text{align}}(y,z) = \left|\mathbb{E}[X \mid Y = y] - \mathbb{E}[X \mid Z = z]\right|^2, \quad c_{\text{oracle}}(y,z) = (y - \text{sgn}(\sigma_{YZ|X})z)^2/2. \tag{10}$$

If $c = c_{\text{oracle}}$, or if $c = c_{\text{align}}$ and $\sigma_{YZ|X}\,\Sigma_{YX}\Sigma_{XZ} > 0$, then there exists a regularization parameter $\lambda^\star > 0$ such that the global bridge in (2) *achieves exact recovery, $Q^\star = P$.*

This result highlights a principled aspect of our dependency-aware bridge: rather than serving merely as an approximate heuristic, it can perfectly reconstruct the unobserved joint Gaussian distribution. The oracle cost c_{oracle} establishes a theoretical guarantee that exact recovery is always possible under Gaussianity. Because this cost depends on the unobserved conditional covariance, it serves primarily as an analytical benchmark. In contrast, the alignment cost c_{align} is fully empirical and computable directly from the separated marginal datasets. The theorem shows that optimizing the bridge with this observable surrogate still achieves exact recovery, provided the latent residual correlation aligns geometrically with the observed marginal projections ($\sigma_{YZ|X}\,\Sigma_{YX}\Sigma_{XZ} > 0$). The explicit forms of the optimal parameter $\lambda^\star$ for both costs, along with the proof, are deferred to Appendix A.

Our recovery analysis provides practical guidance for tuning the hyperparameter λ. If the data is approximately Gaussian and partial information about the joint distribution $P_{Y,Z}$ (e.g., the true correlation or an auxiliary sample) is available, one can directly compute and apply the theoretical $\lambda^\star$. More broadly, if a small validation set of fully observed triplets $(X, Y, Z) \sim P$ is accessible, it can be used to tune λ that maximizes downstream utility, e.g., predictive accuracy, on the completed data.

Assuming access to such auxiliary information is standard practice in the statistical matching literature. For Gaussian population models, Kadane (2001) and Moriarity and Scheuren (2001) assume a prior distribution over σ_{yz} to guide the matching process. In marketing, Gilula et al. (2006) formulated customized dependency structures for binary targets. Kim et al. (2016) leveraged instrumental variable assumptions for Y and Z to recover the full joint via fractional imputation. For categorical data, Fosdick et al. (2016) devised a Bayesian fusion method utilizing an auxiliary data file of (Y, Z) or (X, Y, Z). Conti et al. (2016) restricted the joint distribution by imposing logical constraints on the behavior of Y and Z given X (e.g., total expenditure Y cannot exceed linearly transformed income Z). Moretti and Shlomo (2023) proposed calibrating a prediction-based matching method to account for distribution shifts, justifying the approach under a Gaussian model. These efforts, alongside the works by Singh et al. (1993) and Van Delden et al. (2020), highlight the established necessity of incorporating auxiliary structural information to relax the restrictive CIA paradigm.

4 Experiments

Goal We evaluate whether the proposed Schrödinger bridge (SB) recovers meaningful latent dependence beyond the CIA baseline on Gaussian, synthetic regression, and real-world datasets.

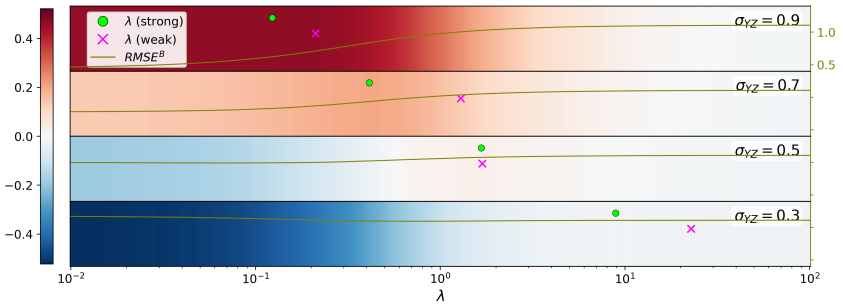


Figure 1: **(Gaussian experiments)** Heatmaps displaying the downstream R^2 improvement ($R^2_{\mathsf{SB}} - R^2_{\mathsf{NAIVE}}$) across the regularization parameter λ (log scale) for varying latent dependence σ_{YZ}. Darker red indicates greater informative gain over the NAIVE (CIA) baseline. Solid curves track the downstream test RMSE (right axis). Circles and crosses mark the average improvement in R^2 out of 10 replicates of λ(Strong), λ(Weak), respectively.

The posterior completion procedure (Section 2.2) merges the two marginal datasets into a fully imputed training set $\hat{D}^T = \hat{D}^A \cup \hat{D}^B$, while all downstream performance is measured on a held-out test set $D^T = (X^T, Y^T, Z^T)$ never seen during bridge learning, completion, or downstream training. Our goal is to verify whether recovering the unknown Y–Z dependence translates into tangible gains in downstream prediction for Y, which indirectly assesses the quality of the recovery. Bridge potentials and downstream predictors are parameterized as MLPs; full details are deferred to Appendix B.1. The source code for reproducing our experiments is publicly available at `https://github.com/kooeunho/SB`.

Reference methods We evaluate three core learning schemes: NAIVE predicts Y from X alone (the CIA baseline); SB predicts Y using the pooled imputed dataset $\hat{D}^T$; and ORACLE trains on fully observed triplets (X, Y, Z) to establish an empirical upper bound. We additionally compare against the optimal transport baseline GO and five mixed-method variants, MM1–MM5, which combine parametric models with hot-deck or nearest-neighbor matching (D'Orazio et al., 2006). We report only the best-performing MM variant per setting. To ensure a fair comparison with GO, which natively requires discrete data, all models are evaluated on binned datasets. This protocol isolates the structural benefits of our EOT formulation over the unregularized OT approach (Section 2.1.1), though our bridge inherently accommodates mixed continuous and discrete variables without compromising computational efficiency. Full details regarding the binning procedure are provided in Appendix B.2.

Choice of λ Following the statistical matching convention (e.g., Moretti and Shlomo (2023); Section 3.2), we assume a small evaluation subset $C = \{(X_i, Y_i, Z_i)\}_{i=1}^{50}$ is available. SB (Weak) selects λ via grid search (100 log-spaced values in $[0.01, 100]$) to maximize downstream performance on C, while SB (Strong) uses a large auxiliary set of size $|D^A| = |D^B| = 10$k. The two thus represent practical and ideal regimes, respectively: when λ selection is insensitive to sample size, their performance should coincide. We denote the chosen values by λ(Weak) and λ(Strong).

4.1 Gaussian experiment

We simulate (X, Y, Z) from a zero-mean trivariate Gaussian distribution, fixing the covariances $\sigma_{XY} = \sigma_{XZ} = 0.5$ and varying the latent dependence $\sigma_{YZ} \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. We generate 30k instances total, allocated equally across D^A, D^B, and the fully observed test set D^T. We apply the empirical alignment cost c_{align} in (10). Figure 1 illustrates how score patterns vary by λ against NAIVE. As σ_{YZ} increases, the optimal λ decreases, confirming that stronger latent Y–Z dependence requires more aggressive coupling. At $\sigma_{YZ} = 0.5$, SB achieves a modest improvement over NAIVE in downstream R^2 (0.273 vs. 0.248), whereas for $\sigma_{YZ} \leq 0.3$, SB performs slightly below NAIVE. However, this pattern shifts dramatically for $\sigma_{YZ} \geq 0.7$.

The score trend of λ(Weak) tracks that of λ(Strong) across σ_{YZ}, validating our methodology even with scarce joint samples. Evaluating the full joint dependence without any Y–Z information is

Table 1: (**Gaussian benchmark**) Down. R^2 denotes the downstream test R^2 for predicting Y; RMSE^B denotes the donor-side imputation RMSE for $\hat{Y}^B$; $\mathcal{W}_2^2$ denotes the squared 2-Wasserstein distance between Gaussian approximations of the completed and true joint distributions. The last row of each σ_{YZ} block reports the selected λ for SB (Strong) and SB (Weak). Boldface indicates the best performance among SB (Weak), GO, and Best MM. A dagger (†) indicates that SB (Strong) performs at least as well as all three. A dot (·) indicates a non-applicable entry. Performance values are reported as mean ± sample standard deviation over 10 random seeds.

σ_{YZ}	Metric	NAIVE	ORACLE	SB (Strong)	SB (Weak)	GO	Best MM
0.9	Down. R^2	0.246 ± 0.008	0.714 ± 0.005	$0.711^{\dagger} \pm 0.005$	$\mathbf{0.708 \pm 0.004}$	0.654 ± 0.035	0.249 ± 0.010 (MM4)
	RMSE^B	·	·	0.620 ± 0.004	0.712 ± 0.006	$\mathbf{0.588 \pm 0.015}$	0.864 ± 0.006 (MM2)
	$\mathcal{W}_2^2$	·	·	$0.016^{\dagger} \pm 0.008$	0.046 ± 0.011	$\mathbf{0.025 \pm 0.007}$	0.352 ± 0.013 (MM2)
	Selected λ	·	·	0.123	0.215	·	·
0.7	Down. R^2	0.247 ± 0.009	0.473 ± 0.007	$0.448^{\dagger} \pm 0.008$	$\mathbf{0.441 \pm 0.007}$	0.348 ± 0.054	0.248 ± 0.008 (MM4)
	RMSE^B	·	·	0.933 ± 0.006	0.972 ± 0.006	$\mathbf{0.805 \pm 0.027}$	0.864 ± 0.006 (MM2)
	$\mathcal{W}_2^2$	·	·	$0.035^{\dagger} \pm 0.012$	$\mathbf{0.057 \pm 0.012}$	0.082 ± 0.012	0.146 ± 0.014 (MM3)
	Selected λ	·	·	0.413	0.599	·	·
0.5	Down. R^2	0.248 ± 0.010	0.318 ± 0.009	$0.273^{\dagger} \pm 0.008$	$\mathbf{0.273 \pm 0.009}$	0.074 ± 0.090	0.250 ± 0.010 (MM4)
	RMSE^B	·	·	1.074 ± 0.007	1.072 ± 0.007	0.974 ± 0.041	$\mathbf{0.864 \pm 0.007}$ (MM2)
	$\mathcal{W}_2^2$	·	·	0.060 ± 0.012	0.058 ± 0.011	0.185 ± 0.014	$\mathbf{0.045 \pm 0.009}$ (MM3)
	Selected λ	·	·	1.668	1.520	·	·
0.3	Down. R^2	0.251 ± 0.009	0.253 ± 0.009	0.231 ± 0.010	0.229 ± 0.011	-0.184 ± 0.107	$\mathbf{0.250 \pm 0.009}$ (MM2)
	RMSE^B	·	·	1.107 ± 0.009	1.106 ± 0.009	1.103 ± 0.048	$\mathbf{0.864 \pm 0.006}$ (MM2)
	$\mathcal{W}_2^2$	·	·	0.052 ± 0.010	0.051 ± 0.010	0.326 ± 0.028	$\mathbf{0.009 \pm 0.005}$ (MM4)
	Selected λ	·	·	8.902	7.391	·	·
0.1	Down. R^2	0.251 ± 0.009	0.275 ± 0.008	0.225 ± 0.009	0.225 ± 0.009	-0.442 ± 0.089	$\mathbf{0.252 \pm 0.010}$ (MM4)
	RMSE^B	·	·	1.110 ± 0.008	1.110 ± 0.008	1.217 ± 0.034	$\mathbf{0.864 \pm 0.006}$ (MM2)
	$\mathcal{W}_2^2$	·	·	0.057 ± 0.008	0.057 ± 0.008	0.506 ± 0.033	$\mathbf{0.021 \pm 0.011}$ (MM5)
	Selected λ	·	·	83.022	83.022	·	·

fundamentally challenging; when only Σ_{YZ} is known (Kadane, 2001), the theoretical $\lambda^\star$ from Theorem 3.2 can be used.

Table 1 evaluates downstream predictive utility (R^2) and donor-side imputation accuracy ($RMSE^B$). In informative regimes ($\sigma_{YZ} \in \{0.7, 0.9\}$), SB (Strong) consistently achieves the highest downstream R^2, closing 88.9% of the gap to the ORACLE upper bound at $\sigma_{YZ} = 0.7$ and approaching 99.4% at $\sigma_{YZ} = 0.9$. SB (Weak) tracks SB (Strong) closely, attaining 85.8% and 98.7% closure respectively. In contrast, GO closes a smaller fraction of the gap (44.7% and 87.2%), and Best MM remains near the NAIVE baseline across all regimes, offering little downstream benefit. On the donor side, GO attains the lowest $RMSE^B$ at both $\sigma_{YZ} = 0.7$ and $\sigma_{YZ} = 0.9$ among SB (Weak), GO, and Best MM. For $\mathcal{W}_2^2$, SB (Strong) performs best at $\sigma_{YZ} \in \{0.7, 0.9\}$, whereas Best MM performs best at $\sigma_{YZ} \leq 0.5$. Among SB (Weak), GO, and Best MM, the lowest values are attained by SB (Weak) at $\sigma_{YZ} = 0.7$ and by GO at $\sigma_{YZ} = 0.9$. For $\sigma_{YZ} < 0.25$, the residual dependence is in fact negative, i.e., $\sigma_{YZ|X} = \sigma_{YZ} - 0.25$, and thus is not captured by the positively aligned cost. Meanwhile, it is shown that the validation procedure of our method finds larger λ as the dependency gets weaker, while GO substantially underperforms under small σ_{YZ}. This highlights that the CIA reference P_ρ anchored in the construction of the bridge $Q^\star$ serves as a safeguard when Y–Z dependency is weak, or even when the loss is misinformative. Table 10 in the appendix considers two more MLE-based strategies: 1) estimating the full Gaussian distribution by using only C and 2) σ_{YZ} by C and the remaining identifiable moments using D^A and D^B, showing slightly higher performance of ours.

4.2 Regression experiments

Next, regression experiments are designed to examine how well SB exploits residual dependence, i.e., the latent association between Y and Z that cannot be explained by X alone. We simulate a standard normal covariate $X = (X_1, X_2)$ and Gaussian noise (e_1, e_2) with variance $\sigma^2_{\mathrm{noise}} \in \{0.1, 0.5\}$ and correlation $\rho_{\mathrm{noise}} \in \{0, 0.3, 0.6, 0.9\}$. We consider the following three data-generating processes:

- SYN1 (Linear): $Y = X_1 + 2X_2 + e_1, \quad Z = 2X_1 + X_2 + e_2.$

Table 2: (**Regression benchmark for $\sigma^2_{\text{noise}} = 0.5$**) Refer to the exposition in Table 1. Results for $\sigma^2_{\text{noise}} = 0.1$ are reported in Appendix Table 12. The full results for $\sigma^2_{\text{noise}} = 0.5$, including $\rho_{\text{noise}} = 0.0$, are reported in Appendix Table 11.

ρ_{noise}	Metric	NAIVE	ORACLE	SB (Strong)	SB (Weak)	GO	Best MM
SYN1 benchmark							
0.9	Down. R^2	0.908±0.001	0.945±0.001	0.945†±0.001	**0.945±0.001**	0.766±0.014	0.909±0.002 (MM2)
	$RMSE^B$	·	·	0.388±0.003	**0.373±0.003**	1.385±0.027	0.707±0.006 (MM2)
	$\mathcal{W}_2^2$	·	·	0.191±0.018	**0.191±0.018**	0.383±0.023	0.213±0.016 (MM2)
0.6	Down. R^2	0.908±0.001	0.924±0.001	0.923†±0.001	**0.923±0.001**	0.735±0.017	0.908±0.002 (MM5)
	$RMSE^B$	·	·	0.766±0.006	0.766±0.006	1.453±0.041	**0.706±0.007 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.211±0.021	0.211±0.021	0.386±0.027	**0.210±0.016 (MM2)**
0.3	Down. R^2	0.908±0.001	0.912±0.001	0.910†±0.001	**0.910±0.001**	0.704±0.021	0.908±0.001 (MM5)
	$RMSE^B$	·	·	0.920±0.007	0.940±0.008	1.518±0.039	**0.706±0.007 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.223±0.023	0.224±0.022	0.393±0.024	**0.211±0.016 (MM2)**
SYN2 benchmark							
0.9	Down. R^2	0.825±0.004	0.934±0.002	0.928†±0.002	**0.928±0.002**	0.271±0.078	0.648±0.016 (MM1)
	$RMSE^B$	·	·	0.567†±0.008	**0.579±0.008**	1.575±0.082	1.594±0.028 (MM2)
	$\mathcal{W}_2^2$	·	·	0.372†±0.034	**0.374±0.034**	0.635±0.042	0.647±0.039 (MM1)
0.6	Down. R^2	0.825±0.004	0.872±0.003	0.864†±0.003	**0.863±0.003**	0.196±0.039	0.647±0.012 (MM1)
	$RMSE^B$	·	·	0.796†±0.009	**0.811±0.008**	1.653±0.036	1.594±0.028 (MM2)
	$\mathcal{W}_2^2$	·	·	0.386†±0.034	**0.389±0.037**	0.664±0.052	0.573±0.033 (MM1)
0.3	Down. R^2	0.824±0.004	0.836±0.004	0.829†±0.004	**0.829±0.004**	0.090±0.062	0.641±0.013 (MM1)
	$RMSE^B$	·	·	0.921†±0.007	**0.921±0.007**	1.744±0.046	1.593±0.028 (MM2)
	$\mathcal{W}_2^2$	·	·	0.404†±0.039	**0.404±0.039**	0.700±0.062	0.576±0.037 (MM1)
SYN3 benchmark							
0.9	Down. R^2	0.715±0.004	0.876±0.002	0.875†±0.002	**0.874±0.001**	0.561±0.053	0.715±0.006 (MM4)
	$RMSE^B$	·	·	0.418±0.005	**0.357±0.005**	0.911±0.036	0.723±0.006 (MM2)
	$\mathcal{W}_2^2$	·	·	0.312±0.049	**0.309±0.049**	0.517±0.066	0.545±0.065 (MM1)
0.6	Down. R^2	0.715±0.003	0.784±0.003	0.777†±0.003	**0.776±0.002**	0.462±0.039	0.712±0.004 (MM4)
	$RMSE^B$	·	·	0.723±0.005	0.751±0.006	1.012±0.033	**0.721±0.007 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.325†±0.026	**0.327±0.029**	0.530±0.062	0.494±0.059 (MM1)
0.3	Down. R^2	0.715±0.003	0.732±0.003	0.722†±0.002	**0.721±0.003**	0.372±0.031	0.713±0.002 (MM4)
	$RMSE^B$	·	·	0.899±0.006	0.894±0.005	1.093±0.032	**0.720±0.008 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.356±0.052	**0.354±0.052**	0.586±0.066	0.479±0.060 (MM1)

- SYN2 (Nonlinear I): $Y = \sin(X_1) + X_2^2 + e_1, \quad Z = \sin(X_1 + X_2) + e_2.$
- SYN3 (Nonlinear II): $Y = \log(1 + e^{X_1}) + X_2 + e_1, \quad Z = \frac{1}{2}\log(1 + e^{X_1}) + X_2^2 + e_2.$

Because X already explains a substantial portion of both Y and Z in these synthetic scenarios, we employ a cost function that specifically targets the residual structure. We first standardize y and z, subtract the empirical conditional mean explained by X, and then compare the remaining residuals:

$$c(x, y, z) = \Big[\Big(\frac{y - \hat{\mu}_Y}{\hat{\sigma}_Y} - \mathbb{E}\Big[\frac{Y - \hat{\mu}_Y}{\hat{\sigma}_Y}\Big|X = x\Big]\Big) - \Big(\frac{z - \hat{\mu}_Z}{\hat{\sigma}_Z} - \mathbb{E}\Big[\frac{Z - \hat{\mu}_Z}{\hat{\sigma}_Z}\Big|X = x\Big]\Big)\Big]^2,$$

where $\hat{\mu}_Y, \hat{\mu}_Z, \hat{\sigma}_Y, \hat{\sigma}_Z$ are the MLE estimates of the Gaussian means and standard deviations, and the conditional mean is estimated by neural networks.

Table 2 reports results across three regression scenarios. The advantage of SB (Weak) over NAIVE grows with the residual dependence ρ_{noise} and is largest in the nonlinear settings, where the covariate X explains less of the joint structure. In SYN1 (linear), NAIVE is already a strong baseline, yet SB (Weak) still recovers most of the remaining gap to ORACLE once $\rho_{\text{noise}} \geq 0.6$ (e.g., 0.923 vs. 0.924 at $\sigma^2_{\text{noise}} = 0.5$, $\rho_{\text{noise}} = 0.6$). In SYN2, where the dependence is harder to capture, SB (Weak) closes a substantial portion of the gap at $\rho_{\text{noise}} = 0.9$ (0.928 vs. ORACLE 0.934, against NAIVE 0.825), and similarly clear gains appear in SYN3. Throughout, GO stays well below NAIVE; Best MM tracks NAIVE closely on SYN1 and SYN3 but falls well below it on SYN2, indicating that neither competitor reliably exploits the residual signal that SB captures. When the residual dependence is weak ($\rho_{\text{noise}} = 0.3$), SB (Weak) performs comparably to NAIVE, reflecting that

Table 3: (**Cost sensitivity of the SYN1, SYN2, and SYN3 benchmarks for $\sigma^2_{\text{noise}} = 0.5$**) Entries are paired performance gaps (residual cost minus alignment cost), reported as mean (standard error) over 10 random seeds. Positive gaps in Down. R^2 and negative gaps in RMSE^B and $\mathcal{W}_2^2$ favor the residual cost. The full results, including $\sigma^2_{\text{noise}} = 0.1$ and $\rho_{\text{noise}} = 0.0$, are reported in Table 13.

σ^2_{noise}	ρ_{noise}	Metric	SYN1		SYN2		SYN3	
			Strong	Weak	Strong	Weak	Strong	Weak
0.5	0.9	Down. R^2	+0.0685 (0.0008)	+0.0654 (0.0084)	+0.1201 (0.0013)	+0.1289 (0.0186)	+0.1744 (0.0017)	+0.1349 (0.0165)
		RMSE^B	-0.5862 (0.0017)	-0.6005 (0.0016)	-0.3743 (0.0030)	-0.3640 (0.0032)	-0.5019 (0.0030)	-0.5358 (0.0027)
		$\mathcal{W}_2^2$	-0.0527 (0.0028)	-0.0523 (0.0032)	-0.0778 (0.0077)	-0.0774 (0.0077)	-0.0759 (0.0084)	-0.0643 (0.0050)
	0.6	Down. R^2	+0.0332 (0.0004)	+0.0307 (0.0067)	+0.0570 (0.0011)	+0.0589 (0.0128)	+0.0980 (0.0013)	+0.0908 (0.0194)
		RMSE^B	-0.2287 (0.0019)	-0.2287 (0.0019)	-0.1635 (0.0028)	-0.1507 (0.0028)	-0.2406 (0.0015)	-0.2149 (0.0017)
		$\mathcal{W}_2^2$	-0.0261 (0.0024)	-0.0261 (0.0024)	-0.0383 (0.0056)	-0.0361 (0.0053)	-0.0404 (0.0078)	-0.0404 (0.0071)
	0.3	Down. R^2	+0.0089 (0.0002)	+0.0066 (0.0023)	+0.0145 (0.0006)	+0.0155 (0.0041)	+0.0258 (0.0005)	+0.0218 (0.0091)
		RMSE^B	-0.0780 (0.0015)	-0.0581 (0.0011)	-0.0472 (0.0013)	-0.0472 (0.0013)	-0.0754 (0.0017)	-0.0800 (0.0018)
		$\mathcal{W}_2^2$	-0.0108 (0.0010)	-0.0094 (0.0012)	-0.0091 (0.0024)	-0.0091 (0.0024)	-0.0160 (0.0045)	-0.0167 (0.0040)

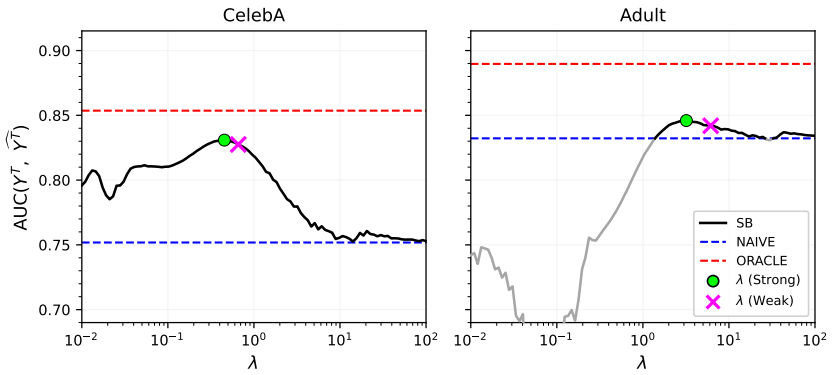


Figure 2: Downstream test AUC, $\text{AUC}(Y^T, \hat{Y}^T)$, vs. regularization parameter λ on CelebA (left) and Adult (right). Dashed lines mark NAIVE (blue) and ORACLE (red); the green dot and magenta cross indicate λ selected by SB (Strong) and SB (Weak), respectively.

little additional predictive information remains to import. SB (Weak) achieves lower $\mathcal{W}_2^2$ than GO and Best MM throughout SYN2 and SYN3. In SYN1, SB achieves the lowest $\mathcal{W}_2^2$ at $\rho_{\text{noise}} = 0.9$, whereas Best MM performs better at lower correlations.

Our guided reconstruction method is sensitive to the choice of the loss function. Table 3 displays the change of evaluation scores, and it shows that the considered regression task favors the residual cost. This tendency is not surprising since the construction of $Q^\star$ relies on the pre-specified loss function. In practice, therefore, the loss function should be carefully designed to reflect the application-specific domain knowledge and improve a target performance utility. In case that a misinformative loss function is chosen by any chance, the CIA baseline anchor becomes activated by choosing a larger λ.

4.3 Real-data experiments

We evaluate SB on two real-world datasets: CelebA (Liu et al., 2015) and Adult (Becker and Kohavi, 1996). For both scenarios, we employ the empirical alignment cost c_{align} defined in (10). For CelebA, we utilize an 8-dimensional VAE latent embedding of $(218, 178, 3)$ images (Kingma and Welling, 2014) as the covariates X, the binary `Attractive` attribute as the target Y, and five binary facial attributes to form a 32-state discrete auxiliary variable Z. For Adult, Y indicates whether income exceeds \$50K, X comprises standard demographic covariates, and Z is constructed via K-means clustering ($K = 32$) over education, occupation, workclass, and financial features.

Consistent with the synthetic results, SB delivers tangible downstream gains on both real-world datasets. On CelebA, as shown in Table 4, SB (Weak) attains the best downstream and donor-side AUC among SB (Weak), GO, and Best MM, closing 73.5% of the downstream NAIVE-to-ORACLE

Table 4: (**Real-data benchmarks**) Down. AUC denotes the downstream test AUC for predicting Y; AUC^B denotes the donor-side imputation AUC for $\hat{Y}^B$. Highlighting conventions and the meaning of $\cdot$ follow Table 1. Values are reported as mean $\pm$ sample standard deviation over 10 random seeds.

Dataset	Metric	NAIVE	ORACLE	SB (Strong)	SB (Weak)	GO	Best MM
CelebA	Down. AUC	0.752 ± 0.002	0.854 ± 0.002	$0.831^{\dagger} \pm 0.002$	$\mathbf{0.827 \pm 0.002}$	0.731 ± 0.013	0.804 ± 0.015 (MM3)
	AUC^B	·	·	$0.832^{\dagger} \pm 0.002$	$\mathbf{0.829 \pm 0.002}$	0.732 ± 0.013	0.793 ± 0.007 (MM5)
Adult	Down. AUC	0.832 ± 0.002	0.890 ± 0.002	$0.846^{\dagger} \pm 0.003$	$\mathbf{0.842 \pm 0.003}$	0.565 ± 0.047	0.832 ± 0.003 (MM1)
	AUC^B	·	·	$0.846^{\dagger} \pm 0.002$	$\mathbf{0.844 \pm 0.002}$	0.568 ± 0.040	0.814 ± 0.005 (MM1)

AUC gap, while GO falls below NAIVE and Best MM trails SB. On Adult, where NAIVE is already strong (0.832), the headroom is small, yet SB (Weak) still leads on both metrics, with a clear donor-side margin (0.844 vs. 0.814). Figure 2 shows that the λ chosen by SB (Weak) using 50 fully observed auxiliary triplets lies near the curve's peak, yielding downstream performance close to SB (Strong). This suggests that a small auxiliary validation set can be sufficient for effective regularization parameter selection.

5 Conclusion

We introduced a dependency-aware Schrödinger bridge for predictive statistical matching. By exponentially tilting the conditional independence baseline with a compatibility cost, our framework preserves observed marginals while actively synthesizing informative latent dependence. Theoretical results guarantee exact joint recovery under Gaussianity, and empirical benchmarks confirm SB maximizes downstream predictive utility when variables exhibit strong underlying correlation.

For practitioners. Our method is most useful when the two datasets share well-harmonized covariates and domain knowledge or auxiliary information supports residual dependence between Y and Z beyond X, as in data recoding. The compatibility cost should encode this knowledge, and a small set of fully observed triplets, when available, can guide the choice of λ and assess predictive gains over the CIA baseline. We recommend checking sensitivity to plausible alternative costs and regularization strengths; a larger λ reduces the influence of the cost and moves the bridge toward CIA. Without credible structural information or joint validation data, the unobserved dependence cannot generally be verified from the marginals alone. In such cases, CIA provides a conservative reference, and dependency-aware reconstructions should be interpreted as sensitivity analyses rather than uniquely identified estimates of the true joint distribution.

Broader impacts and limitations. While dependency-aware matching significantly improves predictive integration, imposing data dependencies risks amplifying historical biases or spurious correlations present in the marginals, necessitating fairness audits when integrating sensitive attributes. Methodologically, our exact recovery guarantees are restricted to Gaussian regimes, and performance depends on the compatibility cost and regularization parameter. While our empirical tuning heuristic performs well, developing a principled, assumption-free tuning criterion remains an open challenge. Future work will explore non-Gaussian recovery, adaptive costs, and algorithmic bias mitigation.

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Appendix

Title: Statistical Matching via Schrödinger Bridge beyond Conditional Independence

A Proofs

A.1 Proposition 2.1

Proof. Let $q(x, y, z)$ and $r(x, y, z)$ denote the densities of Q and R, respectively. The problem (2) seeks to minimize $\mathrm{KL}(Q\|R)$ subject to the marginal constraints $\int q(x, y, z)dz = p_{X,Y}(x, y)$ and $\int q(x, y, z)dy = p_{X,Z}(x, z)$. Introducing Lagrange multipliers $f(x, y)$ and $g(x, z)$ for these constraints, the Lagrangian is given by:

$$\mathcal{L}(q, f, g) = \int q(x, y, z) \log \frac{q(x, y, z)}{r(x, y, z)} dxdydz + \int f(x, y)\Big(p_{X,Y}(x, y) - \int q(x, y, z)dz\Big)dxdy + \int g(x, z)\Big(p_{X,Z}(x, z) - \int q(x, y, z)dy\Big)dxdz.$$

Taking the variational derivative with respect to q and setting it to zero yields the optimality condition:

$$\log \frac{q(x, y, z)}{r(x, y, z)} - f(x, y) - g(x, z) + 1 = 0,$$

which implies $q^\star(x, y, z) = r(x, y, z) \exp(f(x, y) + g(x, z) - 1)$.

By the definition of the reference measure R, $r(x, y, z) \propto e^{-c(x,y,z)/\lambda} p_X(x)p_{Y|X}(y|x)p_{Z|X}(z|x)$. Absorbing the terms depending only on (x, y) into $A(x, y) \propto \exp(f(x, y))$ and those depending only on (x, z) into $B(x, z) \propto \exp(g(x, z))$, we obtain the factorized form:

$$q^\star(x, y, z) = A(x, y)B(x, z)e^{-c(x,y,z)/\lambda}.$$

The integral equations (4) follow by substituting this into the marginal constraints, and the conditional distributions follow by dividing $q^\star(x, y, z)$ by the respective marginals $p_{X,Y}$ and $p_{X,Z}$. □

A.2 Theorem 3.1

Proof. Recall $P = P_{X,Y,Z}$ is the true joint law, $P_\rho = P_X P_{Y|X} P_{Z|X}$ is the CIA product law, and R is defined by $dR(x, y, z) = C_R^{-1} \exp\{-c(x, y, z)/\lambda\} dP_\rho(x, y, z)$, where $C_R = \mathbb{E}_{P_\rho}[\exp\{-c(X, Y, Z)/\lambda\}]$. First, observe that for any joint distribution Q sharing the same (X, Y) and (X, Z) marginals as P, the difference in KL divergences simplifies to:

$$\mathrm{KL}(P\|P_\rho) - \mathrm{KL}(P\|Q) = \mathbb{E}_{P_{X,Y}}[\mathrm{KL}(P_{Z|X,Y}\|P_{Z|X})] - \mathbb{E}_{P_{X,Y}}[\mathrm{KL}(P_{Z|X,Y}\|Q_{Z|X,Y})] = \Delta_Z(Q).$$

Since $Q^\star$ is the KL-projection of R onto $\Pi(P_{X,Y}, P_{X,Z})$, it satisfies the generalized Pythagorean theorem: $\mathrm{KL}(P\|R) \geq \mathrm{KL}(P\|Q^\star) + \mathrm{KL}(Q^\star\|R)$, hence $\mathrm{KL}(P\|Q^\star) \leq \mathrm{KL}(P\|R)$. This implies:

$$\Delta_Z(Q^\star) = \mathrm{KL}(P\|P_\rho) - \mathrm{KL}(P\|Q^\star) \geq \mathrm{KL}(P\|P_\rho) - \mathrm{KL}(P\|R) = \mathbb{E}_P[\log dR] - \mathbb{E}_P[\log dP_\rho].$$

Substituting the definition of dR yields:

$$\Delta_Z(Q^\star) \geq -\log C_R - \frac{1}{\lambda}\mathbb{E}_P[c(X, Y, Z)].$$

Using the inequality $-\log u \geq \frac{1}{2}(u - 1)^2 - (u - 1)$ for all $u \leq 1$, and setting $u = C_R$ (since $c \geq 0$ ensures $C_R \leq 1$), we obtain:

$$\Delta_Z(Q^\star) \geq \frac{1}{2}\left(\mathbb{E}_{P_\rho}[e^{-c(X,Y,Z)/\lambda}] - 1\right)^2 - \left(\mathbb{E}_{P_\rho}[e^{-c(X,Y,Z)/\lambda}] - 1\right) - \frac{1}{\lambda}\mathbb{E}_P[c(X, Y, Z)]$$

$$= C_\lambda - \mathbb{E}_{P_\rho}[e^{-c(X,Y,Z)/\lambda}] + 1 - \frac{1}{\lambda}\mathbb{E}_P[c(X, Y, Z)].$$

Finally, applying the inequality $e^{-v} \le 1 - v + \frac{1}{2}v^2$ for $v = c(X,Y,Z)/\lambda \ge 0$ gives:

$$\Delta_Z(Q^\star) \ge C_\lambda - \mathbb{E}_{P_\rho}\left[1 - \frac{c(X,Y,Z)}{\lambda} + \frac{c(X,Y,Z)^2}{2\lambda^2}\right] + 1 - \frac{1}{\lambda}\mathbb{E}_P[c(X,Y,Z)]$$

$$= C_\lambda + \frac{1}{\lambda}\left(\mathbb{E}_{P_\rho}[c(X,Y,Z)] - \mathbb{E}_P[c(X,Y,Z)]\right) - \frac{1}{2\lambda^2}\mathbb{E}_{P_\rho}[c(X,Y,Z)^2] = C_\lambda + \frac{\delta_c}{\lambda} - \frac{C}{\lambda^2}.$$

This completes the proof. □

A.3 Theorem 3.2

Proof. By the equivalence of the global and local bridge formulations, $Q^\star = P$ if and only if the local bridge $\pi_x^\star$ recovers the true conditional law $P_{Y,Z|X=x}$ for all x. Because the conditional marginals $P_{Y|X=x}$ and $P_{Z|X=x}$ are fixed by the marginal constraints $P_{X,Y}$ and $P_{X,Z}$, any admissible Gaussian coupling π_x is entirely determined by its conditional covariance $q = \mathrm{Cov}_{\pi_x}(Y, Z \mid X = x)$.

Let $\sigma^2_{Y|X}$ and $\sigma^2_{Z|X}$ denote the conditional variance of Y and Z given X. The KL divergence between the coupled Gaussian π_x and the independent baseline ρ_x is:

$$\mathrm{KL}(\pi_x \| \rho_x) = -\frac{1}{2}\log\Big(1 - \frac{q^2}{\sigma^2_{Y|X}\sigma^2_{Z|X}}\Big).$$

The expected cost under π_x takes the form $\mathbb{E}_{\pi_x}[c(Y,Z)] = K - \beta q$, where K depends only on the fixed conditional marginal moments. The cross-term coefficient β depends on the chosen cost:
• For $c = c_{\mathrm{oracle}}$, writing $s = \mathrm{sgn}(\sigma_{YZ|X})$, we have

$$c_{\mathrm{oracle}}(y,z) = \frac{1}{2}(y - sz)^2 = \frac{1}{2}y^2 + \frac{1}{2}z^2 - syz.$$

Since the conditional marginals are fixed, the first two terms contribute only to K, while $\mathbb{E}_{\pi_x}[YZ] = q + \mathbb{E}[Y \mid X = x]\mathbb{E}[Z \mid X = x]$. Hence the q-dependent term is $-sq$, so $\beta = s = \mathrm{sgn}(\sigma_{YZ|X})$.
• For $c = c_{\mathrm{align}}$, the conditional expectations are affine in y and z, with respective linear coefficients $\gamma_Y = \Sigma_{XY}\Sigma_{YY}^{-1}$ and $\gamma_Z = \Sigma_{XZ}\Sigma_{ZZ}^{-1}$. The cross-term of the squared ℓ_2 norm yields $\beta = 2\gamma_Y^\top\gamma_Z$. Because Σ_{YY} and Σ_{ZZ} are positive scalars, we have $\mathrm{sgn}(\beta) = \mathrm{sgn}(\Sigma_{YX}\Sigma_{XZ})$.

The local bridge problem (1) thus minimizes the objective $J(q) = K - \beta q + \lambda \mathrm{KL}(\pi_x \| \rho_x)$. Hence

$$\frac{\partial J}{\partial q} = -\beta + \frac{\lambda q}{\sigma^2_{Y|X}\sigma^2_{Z|X} - q^2} = 0.$$

For exact recovery, we require the optimal covariance to match the true conditional covariance, $q = \sigma_{YZ|X}$. Substituting this into the first-order condition and noting that the denominator becomes the determinant of the true conditional covariance matrix, $|\Sigma_{YZ|X}|$, solving for λ yields

$$\lambda^\star = \frac{\beta|\Sigma_{YZ|X}|}{\sigma_{YZ|X}}.$$

We see that the exact recovery is possible if and only if $\lambda^\star > 0$ if and only if $\mathrm{sgn}(\beta) = \mathrm{sgn}(\sigma_{YZ|X})$.
• If $c = c_{\mathrm{oracle}}$, $\lambda^\star = |\Sigma_{YZ|X}|/|\sigma_{YZ|X}| > 0$ holds universally.
• If $c = c_{\mathrm{align}}$, the condition $\sigma_{YZ|X}\,\Sigma_{YX}\Sigma_{XZ} > 0$ guarantees $\mathrm{sgn}(\beta) = \mathrm{sgn}(\sigma_{YZ|X})$.

Under either condition, setting $\lambda = \lambda^\star$ uniquely minimizes the local bridge objective at the true joint law for all x, resulting in $Q^\star = P$. □

B Experimental Studies

This section consists of the following categories:

- Section B.1 explains the compatibility cost, data preprocessing procedure, and neural network structure in detail for each experiment.
- Section B.2 explains the binning procedure used for continuous variables.
- Section B.3 specifically defines the evaluation metrics.
- Section B.4 introduces the procedure of selecting λ.
- Section B.5 reports the computational resources used for the experiments.

B.1 Architectures, costs, and preprocessing

This section specifies the experimental details. Recall the recipient file is denoted by $D^A = \{(X_i^A, Y_i^A)\}_{i=1}^{n_A}$, the donor file by $D^B = \{(X_j^B, Z_j^B)\}_{j=1}^{n_B}$, and the fully observed test file by $D^T = \{(X_\ell^T, Y_\ell^T, Z_\ell^T)\}_{\ell=1}^{n_T}$. Equivalently, we write

$$D^A = (X^A, Y^A), \qquad D^B = (X^B, Z^B), \qquad D^T = (X^T, Y^T, Z^T).$$

The test file D^T is used only for final evaluation and is not used for bridge learning, posterior completion, downstream training, or selecting λ. For SB (Strong), an additional fully observed auxiliary set of size $|D^A| = |D^B| = 10$k is used for λ-selection only (see Appendix B.4).

The dimensions of X, Y, Z are denoted by d_X, d_Y, d_Z. For a continuous variable W, $\bar{W}$ and s_W denote the empirical mean and empirical standard deviation computed from the file in which W is observed. The standardized value is denoted by

$$W^{\mathrm{std}} = (W - \bar{W})/s_W.$$

Standardization constants are computed from the corresponding observed training file and then applied to validation and test data. Categorical variables are one-hot encoded. A K-state discrete variable is represented either by an index in $\{1, \dots, K\}$ or by the corresponding K-dimensional one-hot vector.

All neural models are implemented in PyTorch. We parameterize the potential $A_p(x, y)$ and the potential $B(x, z)$ as positive multilayer perceptrons with two hidden layers of width 64 and Softplus output activation; the output of $B(x, z)$ is normalized to control its scale. For continuous Gaussian and synthetic experiments, the input dimensions are $d_X + d_Y$ for A_p and $d_X + d_Z$ for B. For real-data experiments, the same hidden-layer design is used after replacing the input dimension by the corresponding latent, one-hot, or discrete auxiliary representation.

The downstream predictor is denoted by f_ψ. It is an MLP with hidden widths 32 and 16 and ReLU activations. Regression experiments use mean squared error. Binary classification experiments use binary cross-entropy. NAIVE trains f_ψ with input X. SB and ORACLE train f_ψ with input (X, Z). The same downstream architecture is used for NAIVE, SB, and ORACLE.

Gaussian experiment. (X, Y, Z) is sampled from a zero-mean trivariate Gaussian distribution with unit variances,

$$\sigma_{XY} = \sigma_{XZ} = 0.5, \qquad \sigma_{YZ} \in \{-0.2, 0, 0.1, 0.25, 0.3, 0.5, 0.7, 0.9\}.$$

Each setting uses 30,000 samples, split equally into D^A, D^B, and D^T (10,000 each). The variables Y^A and Z^B are standardized using their observed training-file values before bridge learning, and the imputed values are transformed back to the original scale for downstream training and evaluation.

For the Gaussian bridge, we use the empirical alignment cost in standardized coordinates while preserving the raw-scale covariance structure. Specifically, if y^{std} and z^{std} denote the standardized versions of y and z, and s_Y, s_Z, s_{XY}, s_{XZ} denote the corresponding empirical standard deviations and covariances, the cost is implemented as

$$c(x, y, z) = \left(s_{XY} s_Y y^{\mathrm{std}} - s_{XZ} s_Z z^{\mathrm{std}}\right)^2.$$

This is the sample-level implementation of the Gaussian alignment cost

$$c_{\mathrm{align}}(y, z) = \|\mathbb{E}[X \mid Y = y] - \mathbb{E}[X \mid Z = z]\|_2^2$$

under the trivariate Gaussian model.

Synthetic regression experiments. $X = (X_1, X_2) \sim N(0, I_2)$, and the noise pair (e_1, e_2) has common variance $\sigma_{\mathrm{noise}}^2 \in \{0.1, 0.5\}$ and correlation $\rho_{\mathrm{noise}} \in \{0, 0.3, 0.6, 0.9\}$. The three data-generating mechanisms are

$$\begin{aligned}
&\mathrm{SYN1}: \quad Y = X_1 + 2X_2 + e_1, \qquad Z = 2X_1 + X_2 + e_2,\\
&\mathrm{SYN2}: \quad Y = \sin(X_1) + X_2^2 + e_1, \qquad Z = \sin(X_1 + X_2) + e_2,\\
&\mathrm{SYN3}: \quad Y = \log(1 + e^{X_1}) + X_2 + e_1, \qquad Z = \tfrac{1}{2}\log(1 + e^{X_1}) + X_2^2 + e_2.
\end{aligned}$$

Each synthetic setting uses 30,000 samples, split equally into D^A, D^B, and D^T.

The cost compares standardized residuals after subtracting the estimated conditional means explained by X. With

$$u_Y(y) = \frac{y - \bar{Y}^A}{s_Y^A}, \qquad u_Z(z) = \frac{z - \bar{Z}^B}{s_Z^B},$$

let $\hat{m}_Y(x)$ be the fitted conditional mean of $u_Y(Y)$ given $X = x$, trained on D^A, and let $\hat{m}_Z(x)$ be the fitted conditional mean of $u_Z(Z)$ given $X = x$, trained on D^B. The residual cost is

$$c(x, y, z) = \frac{\left[\{u_Y(y) - \hat{m}_Y(x)\} - \{u_Z(z) - \hat{m}_Z(x)\}\right]^2}{K_{\mathrm{res}}},$$

where K_{res} is the empirical variance of the numerator over the Monte Carlo triples used for bridge training.

CelebA. X is an 8-dimensional VAE latent embedding, Y is the binary `Attractive` attribute, and Z is the 32-state discrete auxiliary variable induced by the five binary attributes `Young`, `High Cheekbones`, `Arched Eyebrows`, `Heavy Makeup`, and `No Beard`. Equivalently, each Z-state corresponds to one binary vector in $\{0, 1\}^5$, so there are $2^5 = 32$ possible auxiliary states. The dataset is split into 40% for D^A, 40% for D^B, and 20% for D^T.

Adult. Y indicates income above \$50K. The common covariates X consist of age and the categorical variables sex, race, native-country, marital-status, and relationship. The auxiliary variable Z is built from education-num, hours-per-week, capital-gain, capital-loss, education, occupation, and workclass. Numerical variables are standardized, categorical variables are one-hot encoded, and the resulting auxiliary representation is discretized into $K = 32$ clusters by K-means. The split uses 16,000 samples for D^A, 16,000 samples for D^B, and the remainder for D^T.

Training hyperparameters. Unless otherwise stated, the bridge potentials are trained using Adam with learning rate 5×10^{-4}. The bridge mini-batch size is 256, and the Monte Carlo sample size for approximating the inner expectations is 256. We use a scale regularization coefficient 10^{-3}, loss weights $w_A = 2$ and $w_B = 1$, and gradient clipping with maximum norm 5. For posterior completion, each missing value is sampled from a candidate pool of size 1024, i.e., $M_Z = 1024$ and $M_Y = 1024$, unless the corresponding support is finite and directly enumerated. Imputation is performed in batches of size 2000. For downstream prediction, the downstream MLP is trained using Adam with learning rate 10^{-3}, mini-batch size 256, and at most 500 epochs with early stopping patience 30 on a 20% validation split of the downstream training data.

Completion. Each missing Z_i^A is sampled from a candidate pool $\mathcal{C}_Z = \{z_1, \ldots, z_{M_Z}\}$. For an observed pair (x_i^A, y_i^A), the unnormalized posterior weight of candidate z_k is $\omega_{ik}^A = \exp\left\{\log B(x_i^A, z_k) - \frac{c(x_i^A, y_i^A, z_k)}{\lambda}\right\}$, and the normalized probability is $p_{ik}^A = \frac{\omega_{ik}^A}{\sum_{\ell=1}^{M_Z} \omega_{i\ell}^A}$. The completed value $\hat{Z}_i^A$ is sampled from $\mathcal{C}_Z$ according to $\{p_{ik}^A\}_{k=1}^{M_Z}$. For donor-side completion, each missing Y_j^B is sampled from a candidate pool $\mathcal{C}_Y = \{y_1, \ldots, y_{M_Y}\}$. For an observed pair (x_j^B, z_j^B), the unnormalized posterior weight of candidate y_k is

$$\omega_{jk}^B = \exp\left\{\log A_p(x_j^B, y_k) + \log \hat{d}(x_j^B, y_k, z_j^B) - \frac{c(x_j^B, y_k, z_j^B)}{\lambda}\right\}.$$

where $\hat{d}(x, y, z) = \frac{\hat{p}(y|x)}{\hat{p}(z|x)}$. Note that $\hat{p}(z|x)$ is a constant.

Density-ratio estimation. The conditional densities $\hat{p}_{Y|X}$ and $\hat{p}_{Z|X}$ are estimated separately from D^A and D^B, respectively. For the Gaussian experiment, we assume Gaussian conditionals and plug in their closed-form maximum likelihood estimators. For the synthetic regression experiments, we again assume Gaussian conditionals: the conditional means are estimated by neural networks, and the conditional standard deviations are taken as the corresponding residual RMSE. For Adult and CelebA, where Y and Z are categorical, we fit neural classifiers in each marginal and use the output softmax probabilities directly.

Pooled completed data. The completed recipient and donor files are

$$\hat{D}^A = \{(X_i^A, Y_i^A, \hat{Z}_i^A)\}_{i=1}^{n_A}, \qquad \hat{D}^B = \{(X_j^B, \hat{Y}_j^B, Z_j^B)\}_{j=1}^{n_B}.$$

The pooled completed data are denoted by

$$\hat{D}^T = (X', Y', Z') := (X^A \cup X^B,\ Y^A \cup \hat{Y}^B,\ \hat{Z}^A \cup Z^B),$$

on which the downstream predictor f_ψ is trained. For λ, we use the selection procedure described in Appendix B.4.

To see further details, refer to the code scripts in Supplementary.

B.2 Common discrete evaluation protocol

Because GO is defined on a discrete support, Gaussian and synthetic benchmark comparisons are evaluated in a common finite state space. We use the following bin configurations:

$$(n_X, n_Y, n_Z) \in \{(3,3,3), (4,3,4), (5,3,5), (6,3,4), (4,4,4), (4,3,6), (5,4,5), (6,4,6)\}.$$

For a given configuration, n_X is the number of bins used for each coordinate of X, n_Y is the number of bins used for Y, and n_Z is the number of bins used for Z. If X is multivariate, each coordinate is binned separately and the resulting vector of bin labels is used as the discrete X-state.

To avoid leakage, bin boundaries are computed without using D^T. Boundaries for X are computed from the pooled observed X-values in $D^A \cup D^B$. Boundaries for Y are computed only from the observed Y^A values in D^A. Boundaries for Z are computed only from the observed Z^B values in D^B. The same bin maps are applied to completed values and to evaluation-only true values.

For SB recipient-side reconstruction, posterior probabilities are aggregated at the bin level. Let g_Z be the bin map for Z. For a recipient-side sample i, the posterior probability assigned to the k-th Z-bin is

$$\hat{p}_i^A(k) = \sum_{\ell:\, g_Z(z_\ell)=k} p_{i\ell}^A.$$

The predicted Z-bin is

$$\hat{z}_i^A = \arg\max_{k \in \{1,\dots,n_Z\}} \hat{p}_i^A(k).$$

Downstream comparisons use the same binned Z-representation for every method.

The reported Gaussian and synthetic benchmark tables average over the eight bin configurations and also report the best configuration when applicable. For real-data experiments, no additional binning is applied. CelebA uses the intrinsic 32-state auxiliary variable, where each state is one element of $\{0,1\}^5$. Thus, a reconstructed Z-state is counted as correct only when the entire five-dimensional binary attribute vector is recovered correctly. Adult uses the 32-state auxiliary variable obtained by K-means, and a reconstructed Z-state is counted as correct when the predicted cluster index matches the true cluster index. Unless otherwise stated, the uncertainty values reported in the main tables are computed across independent random seeds after fixing the evaluation protocol, and do not represent variability across bin configurations.

B.3 Evaluation metrics

This section defines the metrics used in the figures and tables. For continuous downstream regression in the Gaussian and synthetic experiments, we report the coefficient of determination R^2 and the root mean squared error. Given held-out test samples $\{(X_i^T, Y_i^T, Z_i^T)\}_{i=1}^{n_T}$ and downstream predictions $\hat{Y}_i^T$, the coefficient of determination is

$$R^2 = 1 - \frac{\sum_{i=1}^{n_T}(Y_i^T - \hat{Y}_i^T)^2}{\sum_{i=1}^{n_T}(Y_i^T - \bar{Y}^T)^2}, \qquad \bar{Y}^T = \frac{1}{n_T}\sum_{i=1}^{n_T} Y_i^T.$$

The downstream root mean squared error is

$$\mathrm{RMSE}_{\mathrm{down}} = \left\{ \frac{1}{n_T}\sum_{i=1}^{n_T}(Y_i^T - \hat{Y}_i^T)^2 \right\}^{1/2}.$$

In Figure 1, the heatmap reports the downstream uplift

$$R^2_{\mathrm{SB}} - R^2_{\mathrm{NAIVE}}.$$

For donor-side continuous imputation, we report the RMSE between the true and imputed Y^B values:

$$\mathrm{RMSE}^B = \left\{ \frac{1}{n_B} \sum_{j=1}^{n_B} (Y^B_{j,\mathrm{true}} - \hat{Y}^B_j)^2 \right\}^{1/2}.$$

Here $Y^B_{j,\mathrm{true}}$ is observed only for evaluation in the fully generated or held-out experimental data, not by the imputation algorithm.

To assess joint-distribution recovery, we report the squared 2-Wasserstein distance, $\mathcal{W}_2^2$, between the pooled completed dataset $\hat{D}^T$ and the fully observed test dataset D^T. For the synthetic regression experiments, we randomly sample 500 observations from each dataset without replacement and standardize both samples using the coordinate-wise means and standard deviations of the sampled true test observations. Let $\{\tilde{u}_i\}_{i=1}^{500}$ and $\{\tilde{v}_j\}_{j=1}^{500}$ denote the standardized completed and true joint observations, respectively. We compute

$$\mathcal{W}_2^2 = \min_{\gamma \in \Pi(a,b)} \sum_{i=1}^{500} \sum_{j=1}^{500} \gamma_{ij} \|\tilde{u}_i - \tilde{v}_j\|_2^2, \qquad a = b = \frac{1}{500}\mathbf{1}.$$

This discrete optimal transport problem is solved exactly using `ot.emd2` from the Python Optimal Transport (POT) library. The computation uses the continuous joint observations rather than their binned representations.

For binary real-data experiments, we report ROC-AUC. The downstream AUC, denoted Down. AUC, is computed from the true test labels $Y^T_i \in \{0, 1\}$ and the predicted scores $\hat{s}^T_i$. Equivalently,

$$\mathrm{AUC} = \Pr(\hat{s}^+ > \hat{s}^-) + \frac{1}{2}\Pr(\hat{s}^+ = \hat{s}^-),$$

where $\hat{s}^+$ and $\hat{s}^-$ denote the scores of randomly chosen positive and negative test samples, respectively. For donor-side binary imputation, AUC^B is computed analogously using the true donor-side labels $Y^B_{j,\mathrm{true}}$ and the imputed probability scores

$$\hat{s}^B_j = \Pr(\hat{Y}^B_j = 1 \mid X^B_j, Z^B_j).$$

When we report the fraction of the NAIVE-to-ORACLE gap recovered by SB, we use

$$\mathrm{GapClosure} = \frac{M_{\mathsf{SB}} - M_{\mathsf{NAIVE}}}{M_{\mathsf{ORACLE}} - M_{\mathsf{NAIVE}}},$$

where $M = R^2$ for continuous regression experiments and $M = \mathrm{AUC}$ for binary classification experiments. This quantity is interpreted only when $M_{\mathsf{ORACLE}} > M_{\mathsf{NAIVE}}$.

Unless otherwise stated, reported values are means over 10 independent random seeds, with uncertainty expressed as one sample standard deviation. The cost-sensitivity tables instead report paired mean differences with standard errors, as specified in their captions. Specifically, for metric values $m_1, \ldots, m_{10}$, we report

$$\bar{m} \pm s_m, \qquad \bar{m} = \frac{1}{10}\sum_{r=1}^{10} m_r, \qquad s_m = \left\{ \frac{1}{9}\sum_{r=1}^{10} (m_r - \bar{m})^2 \right\}^{1/2}.$$

B.4 Selecting λ via auxiliary triplets

This section details the validation-based tuning procedure referenced in the main text (Section 4, "Choice of λ"). We assume access to an auxiliary set of fully observed triplets $C = \{(X_i, Y_i, Z_i)\}_{i=1}^{n_C} \sim P_{X,Y,Z}$, disjoint from D^A, D^B, and the held-out test file D^T. The candidate set is $\Lambda = \{10^{-2+4j/99} : j = 0, \ldots, 99\}$, namely 100 log-spaced values in $[0.01, 100]$.

For each $\lambda \in \Lambda$, the bridge potentials are trained on (D^A, D^B), and the learned posterior is used to construct the pooled imputed dataset $\hat{D}^T(\lambda) = \hat{D}^A(\lambda) \cup \hat{D}^B(\lambda)$. A downstream predictor f_ψ^λ is trained on $\hat{D}^T(\lambda)$ to predict Y' from (X', Z'), and evaluated on C using R^2 for regression and AUC for binary classification:

$$\mathrm{Score}_C(\lambda) = \begin{cases} R^2\Big(Y_C, f_\psi^\lambda(X_C, Z_C)\Big) & \text{(regression)}, \\ \mathrm{AUC}\Big(Y_C, f_\psi^\lambda(X_C, Z_C)\Big) & \text{(binary classification)}. \end{cases}$$

The selected value is $\hat{\lambda} = \arg\max_{\lambda \in \Lambda} \mathrm{Score}_C(\lambda)$. The same selection protocol is applied independently within each random seed, and D^T is reserved for final evaluation only.

SB (Weak) vs. SB (Strong). The two variants differ only in the size of C: SB (Weak) uses a small set with $n_C = 50$, reflecting the typical practical regime in statistical matching where only a few fully observed triplets are publicly available; SB (Strong) uses a large set with $n_C = |D^A| = |D^B| = 10$k, serving as an idealized reference. Comparing the two isolates the sensitivity of λ-selection to the size of the auxiliary set.

B.5 Computational resources

All experiments were implemented in PyTorch and run on a GPU-equipped workstation. The workstation had four NVIDIA L4 GPUs, each with approximately 23GB of memory, an Intel Xeon Gold 6526Y CPU with 32 logical CPUs, and 256GB of system memory. The NVIDIA driver version was 560.35.03 and the CUDA version was 12.6. Data preprocessing and metric aggregation were performed on CPUs, while bridge training, posterior completion, and downstream predictor training were performed on GPUs. A single run, consisting of bridge training, posterior completion, downstream predictor training, and final evaluation for one dataset configuration, random seed, and value of λ, took approximately 5 minutes.

C Additional Experimental Results

This section supplements the main experiments with complete mixed-method comparisons, expanded regression results, cost-sensitivity analyses, and a comparison with Gaussian auxiliary-sample baselines.

Complete benchmark results. Tables 5, 6, 7, 8, and 9 report the individual results for MM1–MM5, complementing the metric-wise Best MM summaries in the main text. These results distinguish downstream predictive utility from donor-side imputation accuracy and distributional agreement: the best-performing matching method can differ across metrics.

Expanded regression results. Tables 11 and 12 report the full regression benchmarks at noise variances 0.5 and 0.1, respectively, including the conditionally independent setting $\rho_{\mathrm{noise}} = 0$. The downstream gains of SB generally increase with residual dependence. At zero residual correlation, these gains largely disappear, and SB can perform slightly below NAIVE. Thus, dependency-aware completion is beneficial when informative residual dependence is present and captured by the chosen cost.

Cost sensitivity. Table 13 compares the residual and alignment costs across all three regression benchmarks and both noise levels. Entries are paired score differences computed within each random seed, with standard errors reported in parentheses. The residual cost generally improves performance when residual dependence is present, whereas the differences are small at zero residual correlation. These results support choosing a compatibility cost that reflects the dependence structure relevant to the application.

Comparison with MLEs. We conduct an additional Gaussian experiment to compare the analytical SB coupling with two Gaussian plug-in baselines that exploit a small set of fully observed auxiliary samples. We generate (X, Y, Z) from a zero-mean trivariate Gaussian distribution with covariance

matrix

$$\Sigma = \begin{pmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & \sigma_{YZ} \\ 0.5 & \sigma_{YZ} & 1 \end{pmatrix}, \qquad \sigma_{YZ} \in \{0.9, 0.5\}.$$

For each random seed, we independently generate $n_A = n_B = n_T = 10{,}000$ observations for the recipient, donor, and held-out test datasets, respectively, together with an additional set C of $n_C = 50$ complete triplets. All methods use the same generated datasets within each run.

We consider two Gaussian plug-in baselines. AUX-ONLY estimates the full Gaussian mean vector and covariance matrix using only the 50 complete triplets in C. In contrast, MARGINAL-PLUGIN uses $D^A = (X, Y)$ and $D^B = (X, Z)$ to estimate all identifiable Gaussian moments. Specifically, the mean and variance of X are estimated from the pooled observed X samples, the moments involving (X, Y) are estimated from D^A, and those involving (X, Z) from D^B. Only the unidentified covariance σ_{YZ} is estimated from C.

For SB (Weak), we estimate the Gaussian conditional marginals from D^A and D^B and construct the Gaussian bridge analytically using the alignment cost in (10), without training neural bridge potentials. The regularization parameter is fixed at $\lambda = 0.215$ for $\sigma_{YZ} = 0.9$ and $\lambda = 1.520$ for $\sigma_{YZ} = 0.5$. These values were preselected under a weak-validation setting and are held fixed across random seeds.

For all three methods, the fitted Gaussian conditional distributions provide the completion rules for $Z \mid X, Y$ and $Y \mid X, Z$. We use conditional-mean imputation to construct the pooled training data for downstream prediction and to compute the donor-side imputation error RMSE^B. All methods use the same downstream MLP with two hidden layers of widths 32 and 16, ReLU activations, and Adam optimization. The downstream predictor is evaluated on the fully observed held-out test data.

For distributional evaluation, we instead perform stochastic completion by sampling from the corresponding Gaussian conditional distributions. We randomly select 2,000 observations from each of the completed and true test datasets, standardize both using the means and standard deviations of the sampled true test data, and compute $\mathcal{W}_2^2$ using the closed-form Gaussian 2-Wasserstein distance between their fitted Gaussian approximations. Table 10 reports the mean and sample standard deviation over ten independent random seeds.

In Table 10, the analytical SB construction achieves the best mean scores across the reported metrics, although several improvements over the plug-in baselines are small relative to the reported variability. Because this experiment uses an analytical Gaussian bridge and conditional-mean completion for predictive evaluation, its scores should be interpreted within this comparison rather than directly compared with the main neural-bridge benchmark.

Table 5: **Gaussian benchmark with all mixed-method variants.** Down. R^2 denotes the downstream test R^2 for predicting Y; RMSE^B denotes the donor-side imputation RMSE for $\hat{Y}^B$; $\mathcal{W}_2^2$ denotes the squared 2-Wasserstein distance between Gaussian approximations of the completed and true joint distributions. Boldface indicates the best-performing mixed-method variant for each metric. Values are reported as mean $\pm$ sample standard deviation over 10 random seeds.

σ_{YZ}	Metric	MM1	MM2	MM3	MM4	MM5
0.9	Down. R^2	0.242 ± 0.014	0.247 ± 0.008	0.249 ± 0.016	$\mathbf{0.249 \pm 0.010}$	0.246 ± 0.016
	$RMSE^B$	1.224 ± 0.008	$\mathbf{0.864 \pm 0.006}$	1.220 ± 0.008	1.225 ± 0.007	1.223 ± 0.006
	$\mathcal{W}_2^2$	0.386 ± 0.020	$\mathbf{0.352 \pm 0.013}$	0.382 ± 0.021	0.389 ± 0.018	0.381 ± 0.021
0.7	Down. R^2	0.246 ± 0.015	0.247 ± 0.009	0.245 ± 0.015	$\mathbf{0.248 \pm 0.008}$	0.247 ± 0.009
	$RMSE^B$	1.224 ± 0.010	$\mathbf{0.864 \pm 0.006}$	1.220 ± 0.007	1.226 ± 0.008	1.222 ± 0.007
	$\mathcal{W}_2^2$	0.147 ± 0.012	0.197 ± 0.013	$\mathbf{0.146 \pm 0.014}$	0.150 ± 0.011	0.147 ± 0.010
0.5	Down. R^2	0.248 ± 0.011	0.250 ± 0.010	0.247 ± 0.011	$\mathbf{0.250 \pm 0.010}$	0.248 ± 0.010
	$RMSE^B$	1.224 ± 0.005	$\mathbf{0.864 \pm 0.007}$	1.220 ± 0.007	1.226 ± 0.008	1.221 ± 0.008
	$\mathcal{W}_2^2$	0.049 ± 0.011	0.137 ± 0.010	$\mathbf{0.045 \pm 0.009}$	0.047 ± 0.007	0.046 ± 0.007
0.3	Down. R^2	0.250 ± 0.009	$\mathbf{0.250 \pm 0.009}$	0.250 ± 0.008	0.250 ± 0.009	0.250 ± 0.009
	$RMSE^B$	1.225 ± 0.007	$\mathbf{0.864 \pm 0.006}$	1.220 ± 0.007	1.225 ± 0.008	1.221 ± 0.008
	$\mathcal{W}_2^2$	0.009 ± 0.004	0.126 ± 0.010	0.009 ± 0.004	$\mathbf{0.009 \pm 0.005}$	0.009 ± 0.005
0.25	Down. R^2	0.250 ± 0.009	$\mathbf{0.251 \pm 0.009}$	0.250 ± 0.008	0.250 ± 0.009	0.250 ± 0.009
	$RMSE^B$	1.227 ± 0.010	$\mathbf{0.864 \pm 0.006}$	1.220 ± 0.007	1.225 ± 0.008	1.221 ± 0.008
	$\mathcal{W}_2^2$	0.008 ± 0.006	0.128 ± 0.010	0.008 ± 0.005	$\mathbf{0.007 \pm 0.005}$	0.008 ± 0.006
0.1	Down. R^2	0.249 ± 0.009	0.251 ± 0.009	0.251 ± 0.009	$\mathbf{0.252 \pm 0.010}$	0.248 ± 0.011
	$RMSE^B$	1.222 ± 0.011	$\mathbf{0.864 \pm 0.006}$	1.219 ± 0.007	1.224 ± 0.007	1.221 ± 0.009
	$\mathcal{W}_2^2$	0.022 ± 0.008	0.148 ± 0.009	0.021 ± 0.007	0.021 ± 0.007	$\mathbf{0.021 \pm 0.011}$
0.0	Down. R^2	$\mathbf{0.252 \pm 0.012}$	0.252 ± 0.009	0.251 ± 0.012	0.251 ± 0.010	0.247 ± 0.012
	$RMSE^B$	1.223 ± 0.013	$\mathbf{0.863 \pm 0.005}$	1.219 ± 0.007	1.224 ± 0.007	1.221 ± 0.010
	$\mathcal{W}_2^2$	$\mathbf{0.042 \pm 0.013}$	0.171 ± 0.010	0.045 ± 0.010	0.045 ± 0.010	0.045 ± 0.015
-0.2	Down. R^2	$\mathbf{0.253 \pm 0.017}$	0.251 ± 0.009	0.252 ± 0.012	0.251 ± 0.013	0.248 ± 0.016
	$RMSE^B$	1.221 ± 0.009	$\mathbf{0.863 \pm 0.005}$	1.219 ± 0.008	1.224 ± 0.007	1.221 ± 0.012
	$\mathcal{W}_2^2$	$\mathbf{0.135 \pm 0.015}$	0.249 ± 0.013	0.136 ± 0.016	0.136 ± 0.016	0.136 ± 0.022

Table 6: **SYN1 benchmark with all mixed-method variants.** Down. R^2 denotes the downstream test R^2 for predicting Y; RMSE^B denotes the donor-side imputation RMSE for $\hat{Y}^B$; $\mathcal{W}_2^2$ denotes the squared 2-Wasserstein distance for the completed joint distribution. Boldface indicates the best-performing mixed-method variant for each metric; ties in the mean are resolved by the smaller standard deviation. Values are reported as mean $\pm$ sample standard deviation over 10 random seeds.

σ^2_{noise}	ρ_{noise}	Metric	MM1	MM2	MM3	MM4	MM5
0.5	**0.9**	Down. R^2	0.908 ± 0.002	$\mathbf{0.909 \pm 0.002}$	0.907 ± 0.003	0.908 ± 0.002	0.909 ± 0.002
		$RMSE^B$	1.003 ± 0.009	$\mathbf{0.707 \pm 0.006}$	0.999 ± 0.005	1.001 ± 0.009	0.997 ± 0.007
		$\mathcal{W}_2^2$	0.242 ± 0.020	$\mathbf{0.213 \pm 0.016}$	0.241 ± 0.020	0.243 ± 0.020	0.242 ± 0.023
0.5	**0.6**	Down. R^2	0.908 ± 0.002	0.908 ± 0.001	0.908 ± 0.002	0.908 ± 0.001	$\mathbf{0.908 \pm 0.002}$
		$RMSE^B$	1.003 ± 0.010	$\mathbf{0.706 \pm 0.007}$	0.999 ± 0.006	1.001 ± 0.010	0.997 ± 0.007
		$\mathcal{W}_2^2$	0.226 ± 0.018	$\mathbf{0.210 \pm 0.016}$	0.229 ± 0.019	0.231 ± 0.020	0.230 ± 0.022
0.5	**0.3**	Down. R^2	0.908 ± 0.002	0.908 ± 0.001	0.908 ± 0.001	0.908 ± 0.001	$\mathbf{0.908 \pm 0.001}$
		$RMSE^B$	1.002 ± 0.009	$\mathbf{0.706 \pm 0.007}$	0.998 ± 0.006	1.001 ± 0.010	0.998 ± 0.007
		$\mathcal{W}_2^2$	0.222 ± 0.019	$\mathbf{0.211 \pm 0.016}$	0.225 ± 0.019	0.227 ± 0.020	0.226 ± 0.022
0.5	**0.0**	Down. R^2	0.908 ± 0.002	$\mathbf{0.909 \pm 0.002}$	0.908 ± 0.002	0.908 ± 0.002	0.908 ± 0.002
		$RMSE^B$	1.000 ± 0.011	$\mathbf{0.707 \pm 0.005}$	1.000 ± 0.012	1.002 ± 0.008	1.000 ± 0.009
		$\mathcal{W}_2^2$	0.229 ± 0.016	$\mathbf{0.219 \pm 0.015}$	0.228 ± 0.017	0.228 ± 0.018	0.229 ± 0.015
0.1	**0.9**	Down. R^2	0.980 ± 0.000	$\mathbf{0.980 \pm 0.000}$	0.980 ± 0.000	0.980 ± 0.000	0.980 ± 0.000
		$RMSE^B$	0.449 ± 0.006	$\mathbf{0.316 \pm 0.003}$	0.447 ± 0.002	0.451 ± 0.004	0.448 ± 0.003
		$\mathcal{W}_2^2$	0.163 ± 0.017	$\mathbf{0.155 \pm 0.016}$	0.163 ± 0.016	0.165 ± 0.017	0.165 ± 0.018
0.1	**0.6**	Down. R^2	0.980 ± 0.000	$\mathbf{0.980 \pm 0.000}$	0.980 ± 0.000	0.980 ± 0.000	0.980 ± 0.000
		$RMSE^B$	0.449 ± 0.004	$\mathbf{0.316 \pm 0.003}$	0.447 ± 0.003	0.451 ± 0.004	0.448 ± 0.003
		$\mathcal{W}_2^2$	0.162 ± 0.018	$\mathbf{0.156 \pm 0.016}$	0.162 ± 0.016	0.164 ± 0.017	0.164 ± 0.018
0.1	**0.3**	Down. R^2	0.980 ± 0.000	$\mathbf{0.980 \pm 0.000}$	0.980 ± 0.000	0.980 ± 0.000	0.980 ± 0.000
		$RMSE^B$	0.449 ± 0.004	$\mathbf{0.316 \pm 0.003}$	0.447 ± 0.003	0.451 ± 0.004	0.449 ± 0.003
		$\mathcal{W}_2^2$	0.163 ± 0.018	$\mathbf{0.157 \pm 0.016}$	0.162 ± 0.016	0.164 ± 0.017	0.164 ± 0.018
0.1	**0.0**	Down. R^2	0.980 ± 0.000	$\mathbf{0.980 \pm 0.000}$	0.980 ± 0.000	0.980 ± 0.000	0.980 ± 0.000
		$RMSE^B$	0.449 ± 0.004	$\mathbf{0.316 \pm 0.002}$	0.448 ± 0.005	0.451 ± 0.003	0.449 ± 0.004
		$\mathcal{W}_2^2$	0.165 ± 0.015	$\mathbf{0.161 \pm 0.015}$	0.165 ± 0.015	0.166 ± 0.016	0.166 ± 0.016

Table 7: **SYN2 benchmark with all mixed-method variants.** Down. R^2 denotes the downstream test R^2 for predicting Y; RMSE^B denotes the donor-side imputation RMSE for $\hat{Y}^B$; $\mathcal{W}_2^2$ denotes the squared 2-Wasserstein distance for the completed joint distribution. Boldface indicates the best-performing mixed-method variant for each metric; ties in the mean are resolved by the smaller standard deviation. Values are reported as mean $\pm$ sample standard deviation over 10 random seeds.

σ^2_{noise}	ρ_{noise}	Metric	MM1	MM2	MM3	MM4	MM5
0.5	**0.9**	Down. R^2	$\mathbf{0.648 \pm 0.016}$	0.357 ± 0.026	0.582 ± 0.023	0.575 ± 0.021	0.569 ± 0.022
		$RMSE^B$	2.226 ± 0.033	$\mathbf{1.594 \pm 0.028}$	2.242 ± 0.034	2.268 ± 0.033	2.241 ± 0.035
		$\mathcal{W}_2^2$	$\mathbf{0.647 \pm 0.039}$	0.690 ± 0.052	0.734 ± 0.040	0.705 ± 0.044	0.738 ± 0.058
0.5	**0.6**	Down. R^2	$\mathbf{0.647 \pm 0.012}$	0.356 ± 0.023	0.574 ± 0.015	0.572 ± 0.021	0.570 ± 0.028
		$RMSE^B$	2.246 ± 0.032	$\mathbf{1.594 \pm 0.028}$	2.242 ± 0.034	2.267 ± 0.032	2.242 ± 0.035
		$\mathcal{W}_2^2$	$\mathbf{0.573 \pm 0.033}$	0.662 ± 0.049	0.664 ± 0.040	0.636 ± 0.044	0.669 ± 0.053
0.5	**0.3**	Down. R^2	$\mathbf{0.641 \pm 0.013}$	0.351 ± 0.017	0.576 ± 0.014	0.564 ± 0.025	0.566 ± 0.017
		$RMSE^B$	2.223 ± 0.036	$\mathbf{1.593 \pm 0.028}$	2.242 ± 0.033	2.267 ± 0.032	2.243 ± 0.035
		$\mathcal{W}_2^2$	$\mathbf{0.576 \pm 0.037}$	0.648 ± 0.048	0.629 ± 0.039	0.604 ± 0.044	0.637 ± 0.049
0.5	**0.0**	Down. R^2	$\mathbf{0.643 \pm 0.017}$	0.360 ± 0.012	0.575 ± 0.021	0.574 ± 0.019	0.565 ± 0.025
		$RMSE^B$	2.225 ± 0.038	$\mathbf{1.593 \pm 0.021}$	2.241 ± 0.038	2.267 ± 0.032	2.242 ± 0.032
		$\mathcal{W}_2^2$	$\mathbf{0.538 \pm 0.041}$	0.651 ± 0.059	0.607 ± 0.046	0.603 ± 0.054	0.624 ± 0.056
0.1	**0.9**	Down. R^2	$\mathbf{0.746 \pm 0.020}$	0.408 ± 0.035	0.597 ± 0.028	0.594 ± 0.020	0.599 ± 0.024
		$RMSE^B$	2.039 ± 0.033	$\mathbf{1.463 \pm 0.029}$	2.034 ± 0.036	2.088 ± 0.034	2.030 ± 0.034
		$\mathcal{W}_2^2$	$\mathbf{0.523 \pm 0.037}$	0.602 ± 0.055	0.697 ± 0.038	0.661 ± 0.055	0.702 ± 0.060
0.1	**0.6**	Down. R^2	$\mathbf{0.743 \pm 0.017}$	0.421 ± 0.023	0.596 ± 0.026	0.587 ± 0.027	0.587 ± 0.025
		$RMSE^B$	2.051 ± 0.042	$\mathbf{1.463 \pm 0.029}$	2.035 ± 0.035	2.088 ± 0.033	2.031 ± 0.034
		$\mathcal{W}_2^2$	$\mathbf{0.531 \pm 0.045}$	0.595 ± 0.055	0.674 ± 0.039	0.642 ± 0.055	0.681 ± 0.059
0.1	**0.3**	Down. R^2	$\mathbf{0.746 \pm 0.014}$	0.419 ± 0.027	0.600 ± 0.028	0.582 ± 0.030	0.595 ± 0.024
		$RMSE^B$	2.042 ± 0.044	$\mathbf{1.463 \pm 0.029}$	2.035 ± 0.035	2.088 ± 0.033	2.031 ± 0.034
		$\mathcal{W}_2^2$	$\mathbf{0.517 \pm 0.051}$	0.591 ± 0.055	0.662 ± 0.039	0.632 ± 0.055	0.670 ± 0.060
0.1	**0.0**	Down. R^2	$\mathbf{0.741 \pm 0.014}$	0.419 ± 0.025	0.594 ± 0.031	0.594 ± 0.022	0.595 ± 0.032
		$RMSE^B$	2.068 ± 0.035	$\mathbf{1.463 \pm 0.025}$	2.035 ± 0.036	2.088 ± 0.033	2.031 ± 0.032
		$\mathcal{W}_2^2$	$\mathbf{0.538 \pm 0.055}$	0.593 ± 0.061	0.646 ± 0.042	0.630 ± 0.063	0.660 ± 0.062

Table 8: **SYN3 benchmark with all mixed-method variants.** Down. R^2 denotes the downstream test R^2 for predicting Y; RMSE^B denotes the donor-side imputation RMSE for $\hat{Y}^B$; $\mathcal{W}_2^2$ denotes the squared 2-Wasserstein distance for the completed joint distribution. Boldface indicates the best-performing mixed-method variant for each metric; ties in the mean are resolved by the smaller standard deviation. Values are reported as mean $\pm$ sample standard deviation over 10 random seeds.

σ^2_{noise}	ρ_{noise}	Metric	MM1	MM2	MM3	MM4	MM5
0.5	**0.9**	Down. R^2	0.710 ± 0.008	0.709 ± 0.004	0.713 ± 0.010	$\mathbf{0.715 \pm 0.006}$	0.710 ± 0.005
		$RMSE^B$	1.022 ± 0.005	$\mathbf{0.723 \pm 0.006}$	1.020 ± 0.005	1.021 ± 0.007	1.023 ± 0.010
		$\mathcal{W}_2^2$	$\mathbf{0.545 \pm 0.065}$	0.546 ± 0.047	0.631 ± 0.027	0.599 ± 0.048	0.622 ± 0.042
0.5	**0.6**	Down. R^2	0.710 ± 0.006	0.710 ± 0.005	0.711 ± 0.003	$\mathbf{0.712 \pm 0.004}$	0.708 ± 0.005
		$RMSE^B$	1.020 ± 0.010	$\mathbf{0.721 \pm 0.007}$	1.020 ± 0.007	1.022 ± 0.010	1.026 ± 0.010
		$\mathcal{W}_2^2$	$\mathbf{0.494 \pm 0.059}$	0.525 ± 0.037	0.575 ± 0.036	0.543 ± 0.041	0.581 ± 0.044
0.5	**0.3**	Down. R^2	0.712 ± 0.004	0.709 ± 0.003	0.711 ± 0.003	$\mathbf{0.713 \pm 0.002}$	0.709 ± 0.005
		$RMSE^B$	1.022 ± 0.008	$\mathbf{0.720 \pm 0.008}$	1.017 ± 0.006	1.018 ± 0.008	1.022 ± 0.006
		$\mathcal{W}_2^2$	$\mathbf{0.479 \pm 0.060}$	0.518 ± 0.048	0.559 ± 0.035	0.525 ± 0.043	0.555 ± 0.040
0.5	**0.0**	Down. R^2	0.712 ± 0.006	0.709 ± 0.006	0.712 ± 0.005	$\mathbf{0.712 \pm 0.006}$	0.711 ± 0.005
		$RMSE^B$	1.020 ± 0.008	$\mathbf{0.720 \pm 0.006}$	1.021 ± 0.012	1.020 ± 0.010	1.023 ± 0.011
		$\mathcal{W}_2^2$	$\mathbf{0.495 \pm 0.030}$	0.541 ± 0.041	0.552 ± 0.030	0.543 ± 0.040	0.560 ± 0.024
0.1	**0.9**	Down. R^2	$\mathbf{0.921 \pm 0.002}$	0.916 ± 0.002	0.919 ± 0.002	0.921 ± 0.002	0.914 ± 0.003
		$RMSE^B$	0.492 ± 0.004	$\mathbf{0.348 \pm 0.003}$	0.492 ± 0.003	0.493 ± 0.005	0.510 ± 0.008
		$\mathcal{W}_2^2$	0.462 ± 0.039	$\mathbf{0.457 \pm 0.046}$	0.581 ± 0.039	0.537 ± 0.051	0.573 ± 0.048
0.1	**0.6**	Down. R^2	0.919 ± 0.002	0.916 ± 0.002	0.919 ± 0.002	$\mathbf{0.920 \pm 0.001}$	0.913 ± 0.003
		$RMSE^B$	0.493 ± 0.006	$\mathbf{0.348 \pm 0.004}$	0.492 ± 0.004	0.493 ± 0.006	0.513 ± 0.007
		$\mathcal{W}_2^2$	$\mathbf{0.437 \pm 0.052}$	0.449 ± 0.036	0.566 ± 0.034	0.520 ± 0.044	0.559 ± 0.044
0.1	**0.3**	Down. R^2	0.920 ± 0.001	0.917 ± 0.002	0.920 ± 0.001	$\mathbf{0.920 \pm 0.002}$	0.914 ± 0.004
		$RMSE^B$	0.491 ± 0.007	$\mathbf{0.347 \pm 0.004}$	0.491 ± 0.004	0.491 ± 0.005	0.510 ± 0.008
		$\mathcal{W}_2^2$	0.470 ± 0.057	$\mathbf{0.447 \pm 0.052}$	0.560 ± 0.041	0.512 ± 0.052	0.545 ± 0.045
0.1	**0.0**	Down. R^2	$\mathbf{0.921 \pm 0.002}$	0.916 ± 0.002	0.920 ± 0.002	0.920 ± 0.001	0.918 ± 0.003
		$RMSE^B$	0.489 ± 0.005	$\mathbf{0.347 \pm 0.003}$	0.492 ± 0.005	0.492 ± 0.005	0.506 ± 0.008
		$\mathcal{W}_2^2$	$\mathbf{0.452 \pm 0.047}$	0.466 ± 0.049	0.554 ± 0.034	0.531 ± 0.047	0.564 ± 0.035

Table 9: **Real-data benchmarks with all mixed-method variants.** Down. AUC denotes the downstream test AUC for predicting Y; AUC^B denotes the donor-side imputation AUC for $\hat{Y}^B$. Boldface indicates the best-performing mixed-method variant for each metric. Values are reported as mean $\pm$ sample standard deviation over 10 random seeds.

Dataset	Metric	MM1	MM2	MM3	MM4	MM5
CelebA	Down. AUC	0.751 ± 0.004	0.786 ± 0.016	$\mathbf{0.804 \pm 0.015}$	0.800 ± 0.005	0.801 ± 0.006
	AUC^B	0.719 ± 0.002	0.773 ± 0.015	0.778 ± 0.016	0.586 ± 0.002	$\mathbf{0.793 \pm 0.007}$
Adult	Down. AUC	$\mathbf{0.832 \pm 0.003}$	0.730 ± 0.034	0.812 ± 0.011	0.796 ± 0.014	0.755 ± 0.019
	AUC^B	$\mathbf{0.814 \pm 0.005}$	0.701 ± 0.028	0.700 ± 0.019	0.566 ± 0.007	0.714 ± 0.018

Table 10: **Comparison with Gaussian auxiliary-sample baselines.** Downstream R^2 (Down.R^2, $\uparrow$), donor-side imputation RMSE ($RMSE^B$, $\downarrow$) and squared 2-Wasserstein distance ($\mathcal{W}_2^2$, $\downarrow$) with $n_C = 50$ complete triplets, averaged over ten independent runs (mean $\pm$ standard deviation). AUX-ONLY estimates the full Gaussian of (X, Y, Z) from the complete triplets only, whereas MARGINAL-PLUGIN additionally uses D^A and D^B for the covariance components identifiable from the two marginals. Boldface marks the best result for each metric.

σ_{YZ}	Metric	SB (Weak)	AUX-ONLY	MARGINAL-PLUGIN
0.9	Down.R^2 ($\uparrow$)	$\mathbf{0.804 \pm 0.004}$	0.795 ± 0.012	0.800 ± 0.006
	$RMSE^B$ ($\downarrow$)	$\mathbf{0.438 \pm 0.004}$	0.446 ± 0.010	0.450 ± 0.018
	$\mathcal{W}_2^2$ ($\downarrow$)	$\mathbf{0.014 \pm 0.006}$	0.015 ± 0.006	0.028 ± 0.022
0.5	Down.R^2 ($\uparrow$)	$\mathbf{0.212 \pm 0.012}$	0.186 ± 0.023	0.199 ± 0.030
	$RMSE^B$ ($\downarrow$)	$\mathbf{0.816 \pm 0.005}$	0.833 ± 0.018	0.822 ± 0.013
	$\mathcal{W}_2^2$ ($\downarrow$)	$\mathbf{0.010 \pm 0.006}$	0.027 ± 0.016	0.015 ± 0.012

Table 11: (**Full Regression benchmark**, $\sigma^2_{\text{noise}} = 0.5$) The first row of each ρ_{noise} block reports the selected λ for SB (Strong) and SB (Weak). Metric definitions, highlighting conventions, and summary statistics follow Table 2.

ρ_{noise}	Metric	NAIVE	ORACLE	SB (Strong)	SB (Weak)	GO	Best MM
SYN1 benchmark							
				$\lambda = 0.112$	$\lambda = 0.085$		
0.9	Down. R^2	0.908±0.001	0.945±0.001	0.945†±0.001	**0.945±0.001**	0.766±0.014	0.909±0.002 (MM2)
	$RMSE^B$	·	·	0.388±0.003	**0.373±0.003**	1.385±0.027	0.707±0.006 (MM2)
	$\mathcal{W}_2^2$	·	·	0.191±0.018	**0.191±0.018**	0.383±0.023	0.213±0.016 (MM2)
				$\lambda = 1.048$	$\lambda = 1.048$		
0.6	Down. R^2	0.908±0.001	0.924±0.001	0.923†±0.001	**0.923±0.001**	0.735±0.017	0.908±0.002 (MM5)
	$RMSE^B$	·	·	0.766±0.006	0.766±0.006	1.453±0.041	**0.706±0.007 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.211±0.021	0.211±0.021	0.386±0.027	**0.210±0.016 (MM2)**
				$\lambda = 3.511$	$\lambda = 5.094$		
0.3	Down. R^2	0.908±0.001	0.912±0.001	0.910†±0.001	**0.910±0.001**	0.704±0.021	0.908±0.001 (MM5)
	$RMSE^B$	·	·	0.920±0.007	0.940±0.008	1.518±0.039	**0.706±0.007 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.223±0.023	0.224±0.022	0.393±0.024	**0.211±0.016 (MM2)**
				$\lambda = 32.745$	$\lambda = 8.111$		
0.0	Down. R^2	0.908±0.002	0.908±0.002	0.907±0.002	0.906±0.002	0.672±0.022	**0.909±0.002 (MM2)**
	$RMSE^B$	·	·	0.991±0.007	0.973±0.007	1.580±0.029	**0.707±0.005 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.238±0.018	0.235±0.018	0.409±0.018	**0.219±0.015 (MM2)**
SYN2 benchmark							
				$\lambda = 0.413$	$\lambda = 0.453$		
0.9	Down. R^2	0.825±0.004	0.934±0.002	0.928†±0.002	**0.928±0.002**	0.271±0.078	0.648±0.016 (MM1)
	$RMSE^B$	·	·	0.567†±0.008	**0.579±0.008**	1.575±0.082	1.594±0.028 (MM2)
	$\mathcal{W}_2^2$	·	·	0.372†±0.034	**0.374±0.034**	0.635±0.042	0.647±0.039 (MM1)
				$\lambda = 0.955$	$\lambda = 1.150$		
0.6	Down. R^2	0.825±0.004	0.872±0.003	0.864†±0.003	**0.863±0.003**	0.196±0.039	0.647±0.012 (MM1)
	$RMSE^B$	·	·	0.796†±0.009	**0.811±0.008**	1.653±0.036	1.594±0.028 (MM2)
	$\mathcal{W}_2^2$	·	·	0.386†±0.034	**0.389±0.037**	0.664±0.052	0.573±0.033 (MM1)
				$\lambda = 3.199$	$\lambda = 3.199$		
0.3	Down. R^2	0.824±0.004	0.836±0.004	0.829†±0.004	**0.829±0.004**	0.090±0.062	0.641±0.013 (MM1)
	$RMSE^B$	·	·	0.921†±0.007	**0.921±0.007**	1.744±0.046	1.593±0.028 (MM2)
	$\mathcal{W}_2^2$	·	·	0.404†±0.039	**0.404±0.039**	0.700±0.062	0.576±0.037 (MM1)
				$\lambda = 10.723$	$\lambda = 75.646$		
0.0	Down. R^2	0.826±0.007	0.825±0.007	0.819†±0.007	**0.818±0.008**	−0.012±0.051	0.643±0.017 (MM1)
	$RMSE^B$	·	·	0.962†±0.012	**0.969±0.011**	1.828±0.047	1.593±0.021 (MM2)
	$\mathcal{W}_2^2$	·	·	0.419†±0.023	**0.423±0.022**	0.805±0.046	0.538±0.041 (MM1)
SYN3 benchmark							
				$\lambda = 0.148$	$\lambda = 0.044$		
0.9	Down. R^2	0.715±0.004	0.876±0.002	0.875†±0.002	**0.874±0.001**	0.561±0.053	0.715±0.006 (MM4)
	$RMSE^B$	·	·	0.418±0.005	**0.357±0.005**	0.911±0.036	0.723±0.006 (MM2)
	$\mathcal{W}_2^2$	·	·	0.312±0.049	**0.309±0.049**	0.517±0.066	0.545±0.065 (MM1)
				$\lambda = 0.792$	$\lambda = 1.048$		
0.6	Down. R^2	0.715±0.003	0.784±0.003	0.777†±0.003	**0.776±0.002**	0.462±0.039	0.712±0.004 (MM4)
	$RMSE^B$	·	·	0.723±0.005	0.751±0.006	1.012±0.033	**0.721±0.007 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.325†±0.026	**0.327±0.029**	0.530±0.062	0.494±0.059 (MM1)
				$\lambda = 3.511$	$\lambda = 3.199$		
0.3	Down. R^2	0.715±0.003	0.732±0.003	0.722†±0.002	**0.721±0.003**	0.372±0.031	0.713±0.002 (MM4)
	$RMSE^B$	·	·	0.899±0.006	0.894±0.005	1.093±0.032	**0.720±0.008 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.356±0.052	**0.354±0.052**	0.586±0.066	0.479±0.060 (MM1)
				$\lambda = 27.186$	$\lambda = 18.738$		
0.0	Down. R^2	0.716±0.005	0.716±0.005	0.707±0.007	0.707±0.008	0.183±0.131	**0.712±0.006 (MM4)**
	$RMSE^B$	·	·	0.965±0.011	0.962±0.011	1.236±0.092	**0.720±0.006 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.378±0.026	**0.378±0.025**	0.636±0.109	0.495±0.030 (MM1)

Table 12: (**Full Regression benchmark**, $\sigma^2_{\text{noise}} = 0.1$) The first row of each ρ_{noise} block reports the selected λ for SB (Strong) and SB (Weak). Metric definitions, highlighting conventions, and summary statistics follow Table 2.

ρ_{noise}	Metric	NAIVE	ORACLE	SB (Strong)	SB (Weak)	GO	Best MM
SYN1 benchmark							
0.9				$\lambda = 0.163$	$\lambda = 0.658$		
	Down. R^2	0.980±0.000	0.984±0.000	0.984†±0.000	**0.984±0.000**	0.840±0.016	0.980±0.000 (MM2)
	$RMSE^B$	·	·	0.188†±0.003	**0.265±0.004**	1.220±0.038	0.316±0.003 (MM2)
	$\mathcal{W}_2^2$	·	·	0.153†±0.016	**0.155±0.016**	0.351±0.032	0.155±0.016 (MM2)
0.6				$\lambda = 0.870$	$\lambda = 0.955$		
	Down. R^2	0.980±0.000	0.982±0.000	0.982†±0.000	**0.982±0.000**	0.833±0.020	0.980±0.000 (MM2)
	$RMSE^B$	·	·	0.334±0.003	0.338±0.003	1.231±0.049	**0.316±0.003 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.157±0.016	0.157±0.016	0.349±0.031	**0.156±0.016 (MM2)**
0.3				$\lambda = 2.656$	$\lambda = 8.111$		
	Down. R^2	0.980±0.000	0.980±0.000	0.980†±0.000	**0.980±0.000**	0.831±0.019	0.980±0.000 (MM2)
	$RMSE^B$	·	·	0.405±0.003	0.428±0.003	1.243±0.045	**0.316±0.003 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.162±0.018	0.164±0.018	0.346±0.030	**0.157±0.016 (MM2)**
0.0				$\lambda = 39.442$	$\lambda = 6.734$		
	Down. R^2	0.980±0.000	0.980±0.000	0.980±0.000	0.980±0.000	0.832±0.010	**0.980±0.000 (MM2)**
	$RMSE^B$	·	·	0.443±0.004	0.434±0.004	1.239±0.018	**0.316±0.002 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.168±0.015	0.167±0.015	0.345±0.022	**0.161±0.015 (MM2)**
SYN2 benchmark							
0.9				$\lambda = 0.376$	$\lambda = 0.498$		
	Down. R^2	0.959±0.001	0.978±0.001	0.977†±0.001	**0.977±0.001**	0.453±0.031	0.746±0.020 (MM1)
	$RMSE^B$	·	·	0.270†±0.003	**0.281±0.004**	1.353±0.038	1.463±0.029 (MM2)
	$\mathcal{W}_2^2$	·	·	0.291±0.026	**0.291±0.027**	0.528±0.057	0.523±0.037 (MM1)
0.6				$\lambda = 0.955$	$\lambda = 1.150$		
	Down. R^2	0.959±0.001	0.967±0.001	0.966†±0.001	**0.966±0.001**	0.439±0.032	0.743±0.017 (MM1)
	$RMSE^B$	·	·	0.370†±0.004	**0.376±0.004**	1.363±0.033	1.463±0.029 (MM2)
	$\mathcal{W}_2^2$	·	·	0.295†±0.026	**0.296±0.026**	0.534±0.051	0.531±0.045 (MM1)
0.3				$\lambda = 2.915$	$\lambda = 2.656$		
	Down. R^2	0.959±0.001	0.961±0.001	0.960†±0.001	**0.960±0.001**	0.417±0.040	0.746±0.014 (MM1)
	$RMSE^B$	·	·	0.423±0.003	**0.422±0.003**	1.385±0.035	1.463±0.029 (MM2)
	$\mathcal{W}_2^2$	·	·	0.303±0.029	**0.302±0.029**	0.545±0.051	0.517±0.051 (MM1)
0.0				$\lambda = 27.186$	$\lambda = 11.768$		
	Down. R^2	0.959±0.002	0.959±0.001	0.958†±0.002	**0.958±0.002**	0.422±0.046	0.741±0.014 (MM1)
	$RMSE^B$	·	·	0.445±0.002	**0.444±0.002**	1.385±0.042	1.463±0.025 (MM2)
	$\mathcal{W}_2^2$	·	·	0.316±0.027	**0.315±0.026**	0.545±0.039	0.538±0.055 (MM1)
SYN3 benchmark							
0.9				$\lambda = 0.034$	$\lambda = 0.044$		
	Down. R^2	0.926±0.001	0.958±0.001	0.958†±0.001	**0.957±0.001**	0.789±0.011	0.921±0.002 (MM1)
	$RMSE^B$	·	·	0.161†±0.002	**0.165±0.002**	0.650±0.013	0.348±0.003 (MM2)
	$\mathcal{W}_2^2$	·	·	0.250†±0.038	**0.250±0.038**	0.451±0.045	0.457±0.046 (MM2)
0.6				$\lambda = 1.048$	$\lambda = 2.420$		
	Down. R^2	0.926±0.001	0.939±0.001	0.939†±0.001	**0.936±0.001**	0.771±0.018	0.920±0.001 (MM4)
	$RMSE^B$	·	·	0.338†±0.003	0.378±0.003	0.676±0.020	**0.348±0.004 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.246†±0.027	**0.247±0.027**	0.442±0.039	0.437±0.052 (MM1)
0.3				$\lambda = 3.199$	$\lambda = 3.199$		
	Down. R^2	0.926±0.001	0.929±0.001	0.928†±0.001	**0.928±0.001**	0.755±0.017	0.920±0.002 (MM4)
	$RMSE^B$	·	·	0.405±0.003	0.405±0.003	0.693±0.019	**0.347±0.004 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.266†±0.044	**0.266±0.044**	0.461±0.050	0.447±0.052 (MM2)
0.0				$\lambda = 18.738$	$\lambda = 83.022$		
	Down. R^2	0.926±0.001	0.926±0.002	0.925†±0.002	**0.925±0.001**	0.730±0.010	0.921±0.002 (MM1)
	$RMSE^B$	·	·	0.439±0.004	0.444±0.004	0.728±0.010	**0.347±0.003 (MM2)**
	$\mathcal{W}_2^2$	·	·	0.286±0.026	**0.286±0.027**	0.469±0.048	0.452±0.047 (MM1)

Table 13: Cost sensitivity of the SYN1, SYN2, and SYN3 benchmarks. Entries are paired performance gaps (residual cost minus alignment cost), reported as mean (standard error) over 10 random seeds. Positive gaps in Down. R^2 and negative gaps in RMSE^B and $\mathcal{W}_2^2$ favor the residual cost.

$\sigma^2_{\mathrm{noise}}$	ρ_{noise}	Metric	SYN1		SYN2		SYN3	
			Strong	Weak	Strong	Weak	Strong	Weak
0.5	0.9	Down. R^2	+0.0685 (0.0008)	+0.0654 (0.0084)	+0.1201 (0.0013)	+0.1289 (0.0186)	+0.1744 (0.0017)	+0.1349 (0.0165)
		RMSE^B	-0.5862 (0.0017)	-0.6005 (0.0016)	-0.3743 (0.0030)	-0.3640 (0.0032)	-0.5019 (0.0030)	-0.5358 (0.0027)
		$\mathcal{W}_2^2$	-0.0527 (0.0028)	-0.0523 (0.0032)	-0.0778 (0.0077)	-0.0774 (0.0077)	-0.0759 (0.0084)	-0.0643 (0.0050)
	0.6	Down. R^2	+0.0332 (0.0004)	+0.0307 (0.0067)	+0.0570 (0.0011)	+0.0589 (0.0128)	+0.0980 (0.0013)	+0.0908 (0.0194)
		RMSE^B	-0.2287 (0.0019)	-0.2287 (0.0019)	-0.1635 (0.0028)	-0.1507 (0.0028)	-0.2406 (0.0015)	-0.2149 (0.0017)
		$\mathcal{W}_2^2$	-0.0261 (0.0024)	-0.0261 (0.0024)	-0.0383 (0.0056)	-0.0361 (0.0053)	-0.0404 (0.0078)	-0.0404 (0.0071)
	0.3	Down. R^2	+0.0089 (0.0002)	+0.0066 (0.0023)	+0.0145 (0.0006)	+0.0155 (0.0041)	+0.0258 (0.0005)	+0.0218 (0.0091)
		RMSE^B	-0.0780 (0.0015)	-0.0581 (0.0011)	-0.0472 (0.0013)	-0.0472 (0.0013)	-0.0754 (0.0017)	-0.0800 (0.0018)
		$\mathcal{W}_2^2$	-0.0108 (0.0010)	-0.0094 (0.0012)	-0.0091 (0.0024)	-0.0091 (0.0024)	-0.0160 (0.0045)	-0.0167 (0.0040)
	0.0	Down. R^2	+0.0001 (0.0001)	+0.0011 (0.0013)	-0.0003 (0.0003)	+0.0018 (0.0010)	-0.0000 (0.0007)	+0.0005 (0.0028)
		RMSE^B	-0.0073 (0.0006)	-0.0243 (0.0009)	-0.0089 (0.0007)	-0.0018 (0.0005)	-0.0088 (0.0004)	-0.0119 (0.0006)
		$\mathcal{W}_2^2$	-0.0008 (0.0004)	-0.0040 (0.0009)	-0.0038 (0.0021)	-0.0008 (0.0009)	-0.0017 (0.0014)	-0.0028 (0.0018)
0.1	0.9	Down. R^2	+0.0156 (0.0001)	+0.0142 (0.0018)	+0.0308 (0.0005)	+0.0369 (0.0068)	+0.0497 (0.0008)	+0.0495 (0.0067)
		RMSE^B	-0.2586 (0.0010)	-0.1825 (0.0010)	-0.1738 (0.0008)	-0.1623 (0.0007)	-0.2719 (0.0014)	-0.2695 (0.0011)
		$\mathcal{W}_2^2$	-0.0119 (0.0010)	-0.0108 (0.0013)	-0.0245 (0.0029)	-0.0246 (0.0032)	-0.0247 (0.0035)	-0.0257 (0.0032)
	0.6	Down. R^2	+0.0069 (0.0001)	+0.0068 (0.0015)	+0.0143 (0.0003)	+0.0165 (0.0037)	+0.0254 (0.0004)	+0.0193 (0.0038)
		RMSE^B	-0.1124 (0.0010)	-0.1081 (0.0010)	-0.0740 (0.0008)	-0.0680 (0.0008)	-0.1082 (0.0013)	-0.0691 (0.0008)
		$\mathcal{W}_2^2$	-0.0078 (0.0009)	-0.0072 (0.0010)	-0.0183 (0.0026)	-0.0167 (0.0023)	-0.0113 (0.0019)	-0.0072 (0.0010)
	0.3	Down. R^2	+0.0017 (0.0000)	+0.0010 (0.0004)	+0.0036 (0.0001)	+0.0049 (0.0014)	+0.0065 (0.0002)	+0.0062 (0.0027)
		RMSE^B	-0.0418 (0.0006)	-0.0184 (0.0004)	-0.0217 (0.0005)	-0.0233 (0.0004)	-0.0417 (0.0009)	-0.0417 (0.0009)
		$\mathcal{W}_2^2$	-0.0031 (0.0005)	-0.0008 (0.0006)	-0.0087 (0.0022)	-0.0090 (0.0018)	-0.0038 (0.0010)	-0.0038 (0.0010)
	0.0	Down. R^2	+0.0001 (0.0000)	-0.0004 (0.0004)	-0.0000 (0.0001)	+0.0002 (0.0006)	+0.0000 (0.0002)	+0.0003 (0.0004)
		RMSE^B	-0.0030 (0.0002)	-0.0123 (0.0006)	-0.0014 (0.0002)	-0.0029 (0.0004)	-0.0072 (0.0003)	-0.0016 (0.0002)
		$\mathcal{W}_2^2$	-0.0003 (0.0004)	-0.0007 (0.0004)	-0.0030 (0.0009)	-0.0024 (0.0015)	-0.0008 (0.0006)	-0.0001 (0.0005)

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